Leakage-Free Performance Evaluation of Low-Rate MEMS Sensing for IoT-Based Condition Monitoring of Induction Machines

Hüseyin Tayyer Canseven[a]

[a] *Department of Electrical Engineering, LUT University, Lappeenranta, Finland*

Abstract

Vibration-based condition monitoring on low-cost sensors raises a measurement question that is usually skipped: how should the diagnostic performance of such a sensing chain itself be measured. The question is addressed for a smartphone-class micro-electro-mechanical-systems (MEMS) accelerometer applied to induction-machine monitoring, using a public 100 Hz vibration dataset of six health states, three supply frequencies and two load conditions. Conventional random-split evaluation is shown to overstate accuracy by 41 to 47 percentage points, and a ladder of partitionings then attributes that optimism: window overlap, the mechanism most often blamed and most often guarded against, contributes nothing, as does temporal proximity, while recording identity accounts for all 46.7 points. Enforcing non-overlapping windows would therefore offer no protection on data of this kind. A controlled comparison further shows that the apparent benefit of reducing the sampling rate to 25 Hz is not what it seems: band-limiting to the same frequency range is mildly harmful, the gain follows from the reduced number of samples per analysis window, and omitting the anti-alias filter costs 24.9 points, so a rate sweep with a fixed time-domain feature set confounds the observable band with the sample-count sensitivity of the features. Reported per held-out recording, a resource-frugal configuration—25 Hz, three raw channels and a 4 s window—reaches $75.0 \pm 11.4\%$ against $53.9 \pm 12.5\%$ for the full-rate baseline, a paired improvement of 21.1 points ($p = 0.020$, $n = 36$), while reducing data volume roughly eight-fold. All estimates come from a single machine with one physical specimen per fault, and are reported with intervals clustered on the recording rather than on repeated re-partitionings of the same data.

Keywords: condition monitoring, fault diagnosis, induction machines, MEMS accelerometer, evaluation protocol, data leakage, measurement uncertainty, Internet of Things, low-cost sensing

1. Introduction

Induction machines account for a dominant share of the electrical energy consumed by industry and form the mechanical backbone of countless processes; their unexpected failure causes costly downtime and,

Email address: `huseyin.canseven@lut.fi` (Hüseyin Tayyer Canseven)

in critical applications, safety hazards. Continuous condition monitoring (CM) and early fault diagnosis are therefore central to modern maintenance practice [1, 2]. Among the available modalities, vibration analysis is the most established, because the mechanical and electrical faults of interest—bearing defects, rotor asymmetries, and supply imbalances—imprint characteristic signatures on the machine's vibration response [3, 4].

The prevailing paradigm relies on industrial-grade piezoelectric accelerometers sampled at tens to hundreds of kilohertz, which resolve the high-frequency bearing defect frequencies exploited by spectral and envelope analysis. This is reflected in the public benchmarks that anchor the field, acquired at 8–200 kHz [5, 6, 7, 8]. Although accurate, such instrumentation is costly, requires intrusive installation, and scales poorly to the vast population of small, low-value machines that individually do not justify a dedicated monitoring channel yet collectively dominate a plant's machine inventory. Closing this coverage gap motivates a shift toward inexpensive, wireless, and readily deployable sensing.

Consumer micro-electro-mechanical-systems (MEMS) inertial sensors, epitomised by those embedded in smartphones, offer such a platform at negligible marginal cost [9]. Machine fault detection has been demonstrated from smartphone- and MEMS-acquired vibration [10, 11, 12], and dedicated low-cost MEMS nodes now integrate on-board processing with wireless telemetry for machine diagnostics [13, 14]. These sensors, however, operate at sampling rates one to three orders of magnitude below industrial practice—100 Hz in the present case—so that, by the Nyquist criterion [15, 16], the high-frequency signatures underpinning classical bearing diagnosis are not observable. A fundamental and largely unresolved question follows: what fault information is genuinely recoverable from such a constrained sensor, and does the apparent performance transfer to a real deployment?

The second half of that question is easily overlooked. Learning-based fault diagnosis routinely reports near-perfect accuracy, yet a growing body of evidence attributes much of this success to *data leakage*—information shared between training and test partitions that inflates measured accuracy without improving generalisation [17]. In vibration diagnosis, leakage arises when overlapping analysis windows from a single continuous recording are divided across partitions, or when recording-specific characteristics are learned in place of fault features; enforcing component- or recording-wise separation has been shown to reduce reported accuracy substantially [18, 19]. This scrutiny has so far been directed at industrial-grade, high-rate data. For low-rate smartphone datasets—in which each operating condition is captured in one continuous acquisition and heavy window overlap is common—the risk is arguably greater, yet random-split evaluation remains the norm, including in the baseline accompanying the dataset examined here [20].

A third consideration is deployment. The Industrial Internet of Things envisions dense networks of inexpensive nodes that stream, or locally distil, diagnostic information over constrained wireless links [21], and recent TinyML implementations have shown that lightweight classifiers can run on microcontrollers at millisecond latency and sub-millijoule energy [22, 23]. Such work establishes that edge diagnosis is feasible, but it generally presupposes industrial-bandwidth sensing and does not characterise the coupled trade-off between diagnostic accuracy and the sampling, communication, and computational cost that governs an ultra-low-rate node. Whether, for a smartphone-class sensor, the low sampling rate is a liability to be tolerated or an asset to be exploited has not been quantified.

This paper approaches these gaps as a measurement problem. Using a public smartphone-acquired vibration dataset of an induction machine recorded across six health states, three supply frequencies and two load conditions at 100 Hz [20, 24], and rather than proposing a new classifier, the study treats diagnostic performance as a quantity to be measured: a protocol is developed for estimating it without leakage, the mechanisms that inflate it are isolated experimentally, the accuracy–resource trade-off that governs an IoT node is mapped, and every figure is reported with an interval whose sampling unit is stated. The scope is deliberately bounded, and the bound is stated at the outset: all recordings come from one 1.1 kW machine in which each fault class is represented by a single physical specimen, so what is estimated is performance on an unseen recording of that machine, not performance on an unseen machine.

The principal contributions are fourfold. First, the study *decomposes* the optimism of conventional evaluation into its mechanisms. Random splitting of windows is shown to overstate accuracy by 41 to 47 percentage points, and a ladder of partitionings then attributes that gap: window overlap, the mechanism most often blamed and most often guarded against, accounts for none of it, as does simple temporal proximity, while recording identity accounts for all 46.7 points. Enforcing non-overlapping windows, a common precaution, is therefore no protection at all on data of this kind. Second, it resolves what a sampling-rate sweep actually measures. Decimation to 25 Hz improves leakage-free accuracy even though it removes every shaft rotational frequency in the dataset, and a controlled comparison shows that band-limiting alone is mildly harmful, that the improvement follows from the reduced number of samples per window, and that omitting the anti-alias filter is severely damaging; a rate sweep performed with a fixed time-domain feature set thus confounds the observable band with the sample-count sensitivity of the features. Third, it provides a data-derived characterisation of the smartphone MEMS channel—its observed output increment, an in-operation signal floor, the band made observable by the sampling rate, and the relative informativeness of the raw and gravity-compensated triads—while stating explicitly that these are properties of the delivered signal and not a metrological calibration of the transducer. Finally, it maps the accuracy–resource trade-off into deployable terms through a payload-rate comparison and an edge-model footprint, and reports all comparisons with recording-clustered intervals and paired permutation tests rather than with statistics computed over repeated re-partitionings of the same data.

The remainder of this paper is organised as follows. Section 2 reviews related work on vibration-based condition monitoring, low-cost MEMS sensing, IoT and edge diagnosis, and evaluation methodology. Section 3 describes the dataset and measurement setup, and Section 4 characterises the sensor and its signals. Section 5 details the processing chain and the evaluation protocols. Section 6 reports the results, Section 7 discusses their implications and limitations, and Section 8 concludes.

2. Related work

2.1. Vibration-based condition monitoring of induction machines

Vibration analysis is the most established modality for detecting mechanical and electrical faults in induction machines, covering bearing defects, rotor asymmetries and supply imbalance [1, 2]. Classical pipelines rely on industrial-grade piezoelectric accelerometers sampled at tens to hundreds of kilohertz, which resolve the characteristic bearing defect frequencies—ball-pass frequency outer race (BPFO), ball-pass frequency inner race (BPFI) and ball-spin frequency (BSF)—and their harmonics through spectral and envelope analysis. The signal models underlying that practice, and the decomposition methods built on them, remain an active subject in their own right [25, 26]. Vibration is not the only option: stator-current analysis is attractive because it requires no additional transducer, and comparative studies on induction machines and railway traction drives report that the two modalities are complementary, with vibration generally stronger for mechanical defects and current for electrical asymmetries [27, 28]. Deep networks have since become the default classifier family for both [3, 4, 29, 30].

This paradigm is reflected in the public benchmarks that anchor the field, acquired at 8–16 kHz [5], 42 kHz [6], 51.2 kHz [7] and up to 200 kHz [8]. High-rate acquisition, however, entails costly instrumentation, intrusive cabling and limited portability, which restricts deployment to the comparatively small population of critical machines and leaves the many low-value machines in a plant unmonitored.

2.2. Low-cost MEMS accelerometers as measurement instruments

Micro-electro-mechanical-systems (MEMS) accelerometers are the obvious candidate for closing that coverage gap, and their metrological behaviour has consequently become a subject of measurement research in its own right. The quantities that matter are well identified: sensitivity and its traceability to a reference standard [31], amplitude linearity and temperature dependence under dynamic excitation [32], cross-axis sensitivity [33, 34], thermal drift and hysteresis [35, 36, 37], and the intrinsic noise of the transducer and its readout [38]. Calibration procedures have been developed that reduce the equipment burden, from six-position schemes [39] to in-field methods that estimate cross-axis terms without external references [33] and low-cost thermal compensation that avoids a climatic chamber [37]. Systematic characterisations of consumer-grade devices report performance that is adequate for many vibration tasks but markedly inferior to reference-grade transducers in noise and bandwidth [40, 9], and the propagation of these sensor-level limits into diagnostic features has been examined explicitly [41].

Two findings from this literature bear directly on the present study. First, mounting is a first-order influence quantity: the position and attachment of a MEMS accelerometer measurably changes the vibration it reports [42], which bounds how far any single-installation result can be generalised. Second, low-cost devices are increasingly deployed as measurement instruments in their own right, in structural and machine monitoring, once their limitations are acknowledged and compensated [43, 44]. Condition monitoring practitioners have nonetheless generally regarded consumer-grade MEMS bandwidth as insufficient for bearing diagnosis [13], and purpose-built low-cost nodes therefore target kilohertz rates [14, 45].

2.3. Smartphone-grade sensing and its measurement limitations

Smartphones embed MEMS inertial sensors at negligible marginal cost, and their error characteristics have been measured against laboratory references: bias, scale-factor and noise behaviour differ appreciably between handsets and are sensitive to acquisition settings [46]. Despite this, smartphone acquisition has been used successfully for road-profile estimation [47], structural vibration monitoring with time-synchronised handsets [48] and even submillimetre acoustic vibration measurement [49]. In machine diagnosis specifically, fault detection has been demonstrated from smartphone- and MEMS-acquired vibration using mobile applications and, more recently, deep networks [10, 11, 12]. A recurring constraint is the sampling rate: application-level acquisition is frequently capped at or near 100 Hz, one to three orders of magnitude below industrial practice, so that by the Nyquist criterion the high-frequency signatures underpinning classical bearing diagnosis are not observable [15]. What such a constrained channel can still convey has not been established as a measurement question, and the effect of deliberate rate reduction on diagnostic features has been examined only in passing [50].

The dataset employed in this study [20, 24] extends this line by providing long-duration, multi-condition smartphone-MEMS recordings at a software-capped 100 Hz. Crucially, the accompanying baseline reports near-perfect classification accuracy under a random split of overlapping windows, a protocol that is scrutinised in Section 2.5.

2.4. Networked and edge-based condition monitoring

The Industrial Internet of Things reframes condition monitoring as a distributed sensing problem in which many inexpensive nodes stream, or locally distil, diagnostic information over constrained wireless links [21, 51]. Because raw high-rate vibration cannot be sustained over low-power wide-area radios, a large fraction of recent work pushes feature extraction or inference onto the node. TinyML implementations of lightweight classifiers on microcontrollers report millisecond-scale latency and sub-millijoule energy per inference [22, 23, 52], and embedded sensor nodes now combine on-board learning with prognostic output [53]. On the acquisition side, self-powered wireless vibration nodes [54] and synchronised long-range nodes for modal identification [55] demonstrate that the sensing, energy and communication budgets are tightly coupled. These studies establish the feasibility of edge diagnosis, but they typically assume industrial-bandwidth sensing and rarely quantify the trade-off between sampling and communication cost on one side and diagnostic accuracy on the other, which is precisely what governs an ultra-low-rate node.

2.5. Evaluation methodology, leakage and condition shift

A recurring threat to the credibility of learning-based fault diagnosis is data leakage—the inadvertent sharing of information between training and test partitions that inflates reported accuracy without improving generalisation [17, 56]. In vibration diagnosis two mechanisms dominate: splitting *overlapping* analysis windows drawn from the same continuous recording, so that near-duplicate segments populate both partitions; and recording- or component-wise correlations that let a model recognise *which recording* a window originates from rather than the fault itself. Enforcing component-wise or recording-wise separation has been shown to reduce headline accuracies substantially [18, 19], and the composition of the training set has been shown to determine what a model generalises to at all [57]. The same lesson has been learned independently in wearable sensing, where leave-one-subject-out evaluation replaced random splitting once the latter was found to be optimistic [58]. A related and less frequently addressed bias arises when the same cross-validation scores are used both to select a configuration and to report its performance; unbiased estimation requires the selection to be nested inside the evaluation [59]. Recent work on deep bearing diagnosis argues that benchmark results can collapse to dataset-dependent behaviour that does not transfer [60].

A second and complementary body of work addresses operating-condition shift, in which training and deployment differ in speed, load or installation. Transfer and domain-adaptation methods are the usual response [61, 62, 63, 64], and their prominence is itself evidence that condition shift degrades performance materially. Measuring that degradation directly, rather than attempting to remove it, is the approach taken here. Despite these warnings, random-split evaluation remains prevalent in smartphone- and MEMS-based machine diagnosis, including the baseline accompanying the dataset used here [20].

2.6. Reporting performance with uncertainty

Measurement practice requires that a reported quantity carry a statement of uncertainty, and the framework for doing so is standardised [65]. Type A evaluation from repeated observations is well developed, including its inverse problem [66], its behaviour when random noise and systematic effects act simultaneously [67], and its estimation from a single measurement through propagation [68]. Extending this discipline to quantities produced by learned models is comparatively recent: methodologies have been proposed for GUM-conformant uncertainty evaluation of machine-learning-based measurement models [69], and for making decisions in the presence of measurement uncertainty [70]. Diagnostic accuracy, by contrast, is still routinely reported in the fault-diagnosis literature as a bare point estimate, without an interval, without a statement of the unit over which it was replicated, and often without the information needed to reproduce the partitioning that produced it. That gap between measurement practice and diagnostic reporting practice motivates the protocol developed in Section 5.

2.7. Positioning

Table 1 situates this work against representative prior art. Existing studies advance one axis at a time. High-rate benchmarks and deep models maximise accuracy but ignore cost and evaluation rigour; methodology papers expose leakage but on industrial-grade data; MEMS metrology papers characterise the sensor thoroughly but stop short of the diagnostic task; smartphone-based studies embrace low-cost sensing but retain optimistic random-split evaluation; and IoT/TinyML works demonstrate edge deployment yet assume industrial bandwidth. To the best of the authors' knowledge, no prior study jointly evaluates ultra-low-rate smartphone-MEMS machine diagnosis under a leakage-free, cross-condition protocol, decomposes the resulting optimism into its mechanisms, and maps the accuracy–resource trade-off into communication and edge-model terms. This paper addresses that combination, and reports the result with an interval whose sampling unit is stated.

3. Dataset and Measurement Setup

The present study is based on the publicly available smartphone-acquired vibration dataset introduced in [20] and archived on Mendeley Data [24]. The dataset was selected because it embodies exactly the low-cost, low-bandwidth acquisition regime targeted in this work. This section summarises the test rig, the acquisition chain, the emulated fault conditions, and the organisation of the recordings; the reader is referred to the original data article for the complete experimental protocol. The acquisition and dataset parameters are consolidated in Table 2.

3.1. Test Rig and Machine

The measurements were carried out on a 1.1 kW three-phase squirrel-cage induction machine, a rating representative of the large population of small general-purpose drives found in industry. Variable-speed operation was obtained through a voltage/frequency (V/f)-controlled inverter, and a mechanical load was applied by a DC generator coupled directly to the machine shaft. The generator fed an adjustable resistive load bank, monitored with a digital multimeter, so that the load level could be set to give both loaded and unloaded operation. A consumer smartphone was affixed to the machine terminal box with double-sided adhesive tape, oriented approximately parallel to the shaft axis; the mounting location was marked so that the placement could be reproduced across successive installations. The complete arrangement—induction machine, V/f inverter, coupled DC generator, resistive load bank, and supply-monitoring instrumentation—is shown in Fig. 1.

3.2. Acquisition Chain and Signal Channels

Vibration signals were acquired with an iPhone 15 Pro Max using its integrated tri-axial MEMS inertial sensor, accessed through the *Sensor Play* application. Six channels were recorded for every operating condition: the three raw acceleration components (g_x, g_y, g_z) and the three gravity-compensated linear-acceleration components (g_{userx}, g_{usery}, g_{userz}) produced by the device's on-board sensor fusion. All channels were sampled at 100 Hz, which corresponds to the maximum stable rate supported by the acquisition application, and each operating condition was recorded continuously for 15 minutes. The resulting sampling rate is one to three orders of magnitude below that of the industrial-grade benchmarks discussed in Section 2, and constrains the analysable band to below 50 Hz in accordance with the Nyquist criterion [15, 16].

3.3. Health States and Fault Emulation

Six machine health states were considered: one healthy (fault-free) baseline and five fault conditions spanning mechanical and electrical origins, as summarised in Table 3. The bearing faults were introduced by substituting a 6205ZZ bearing that had been extracted from industrially operated motors, so that the degradation reflects realistic in-service mechanisms rather than artificial machining. Two lubrication-starvation severities were distinguished: insufficient lubrication (B1) and severe insufficient lubrication (B2), the severity being assigned qualitatively from the rotational behaviour observed during manual inspection; a bearing with a cracked outer ring (B3) constituted a localised mechanical defect. The electrical fault, voltage imbalance (V), was produced by inserting voltage-divider resistors in one supply phase, which yielded a pronounced phase-current asymmetry (for example, phase currents of 0.75/2.75/3.03 A at 50 Hz under load [20]). The rotor fault, broken rotor bar (R), was emulated by drilling four holes of 5–9 mm diameter and 16–18 mm depth into the rotor bars. Because fault-related vibration signatures are speed-dependent, the shaft speed was recorded for every condition; the measured values are reproduced in Table 4, where the reduced speed under voltage imbalance (e.g., 1326 r/min versus 1434 r/min for the healthy machine at 50 Hz under load) is consistent with the torque disturbance introduced by that fault.

3.4. Operating Conditions and Dataset Organisation

Each of the six health states was recorded at three supply frequencies (30, 40, and 50 Hz) and under two load conditions (loaded and unloaded), producing a fully crossed matrix of $6 \times 3 \times 2 = 36$ recordings. The files follow the naming convention `Class_Load_SpeedHz`, where the load indicator is `1` for loaded and `0` for unloaded operation; for instance, `B1_1_40Hz` denotes insufficient lubrication under load at 40 Hz. Each recording is provided in both comma-separated (`.csv`) and MATLAB (`.mat`) formats. Representative gravity-compensated waveforms for the six health states, recorded under identical operating conditions, are shown in Fig. 2; the healthy machine exhibits markedly lower vibration amplitude, whereas the fault conditions produce larger and morphologically distinct responses.

3.5. Data Integrity

Prior to analysis, an independent integrity check was performed on all 36 recordings. Each file was confirmed to contain exactly 89,999 samples per channel, corresponding to 900 s at 100 Hz, with no missing

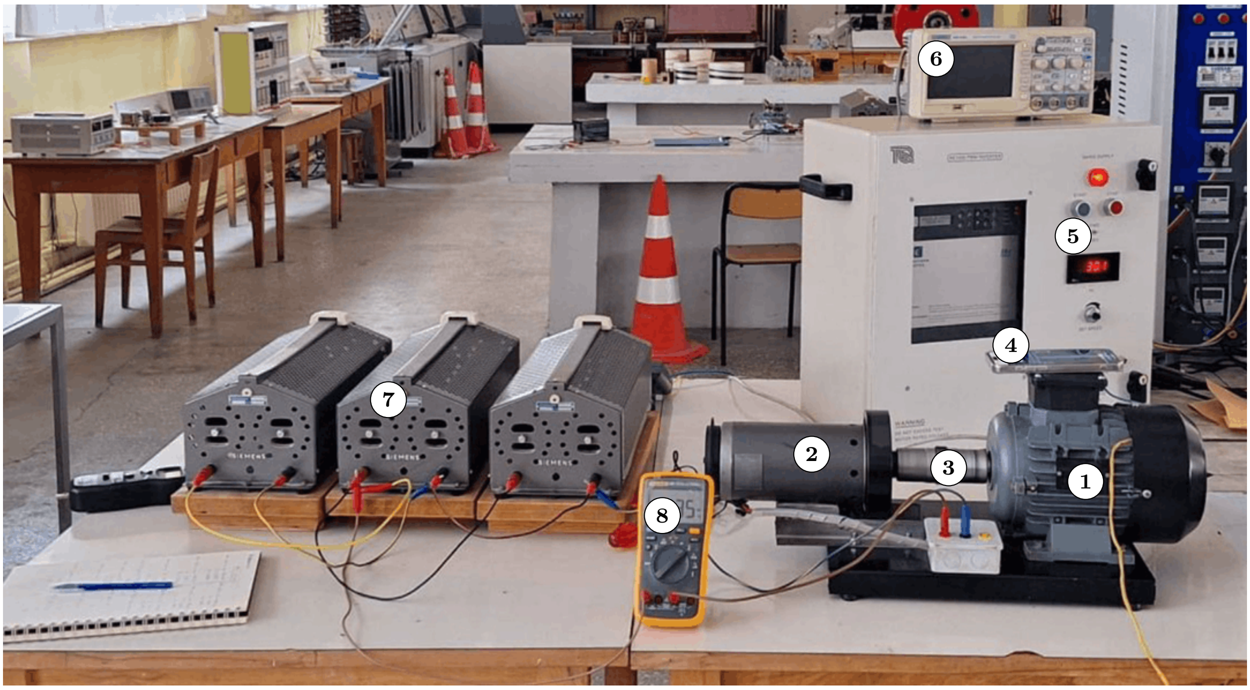

Figure 1: Test-rig arrangement (photograph adapted from [20]): (1) squirrel-cage induction machine under test; (2) shaft-coupled DC generator providing the mechanical load; (3) flexible shaft coupling; (4) smartphone supplying the tri-axial MEMS vibration signal; (5) V/f variable-frequency supply; (6) oscilloscope; (7) resistive load bank; (8) digital multimeter.

values and no systematic sample-and-hold duplication (the maximum fraction of consecutive duplicated rows in any recording was below 0.01%). The channel set and recording length were therefore uniform across the entire dataset, and no imputation or resampling was required.

4. Data-derived characterisation of the signal channels

Before any classifier is trained, it is instructive to establish what fault-related information the recorded channels can, in principle, convey. It is important to be precise about what follows and what does not. A metrological characterisation of a MEMS accelerometer establishes sensitivity traceable to a reference standard, amplitude linearity and frequency response, cross-axis sensitivity, and thermal behaviour, and it requires controlled excitation and reference instrumentation [31, 32, 39, 33, 35, 37]. No such experiments are possible on an archived dataset, and none are claimed here.

What is presented instead is a *data-derived characterisation of the signal channels*: quantities computed directly from the delivered samples that bound what those channels can carry. Three are reported—the observed minimum output increment, an in-operation broadband power spectral density (PSD) floor, and the band made observable by the sampling rate—together with the relative diagnostic value of the raw and gravity-compensated triads. These are properties of the delivered signal, not of the transducer: each

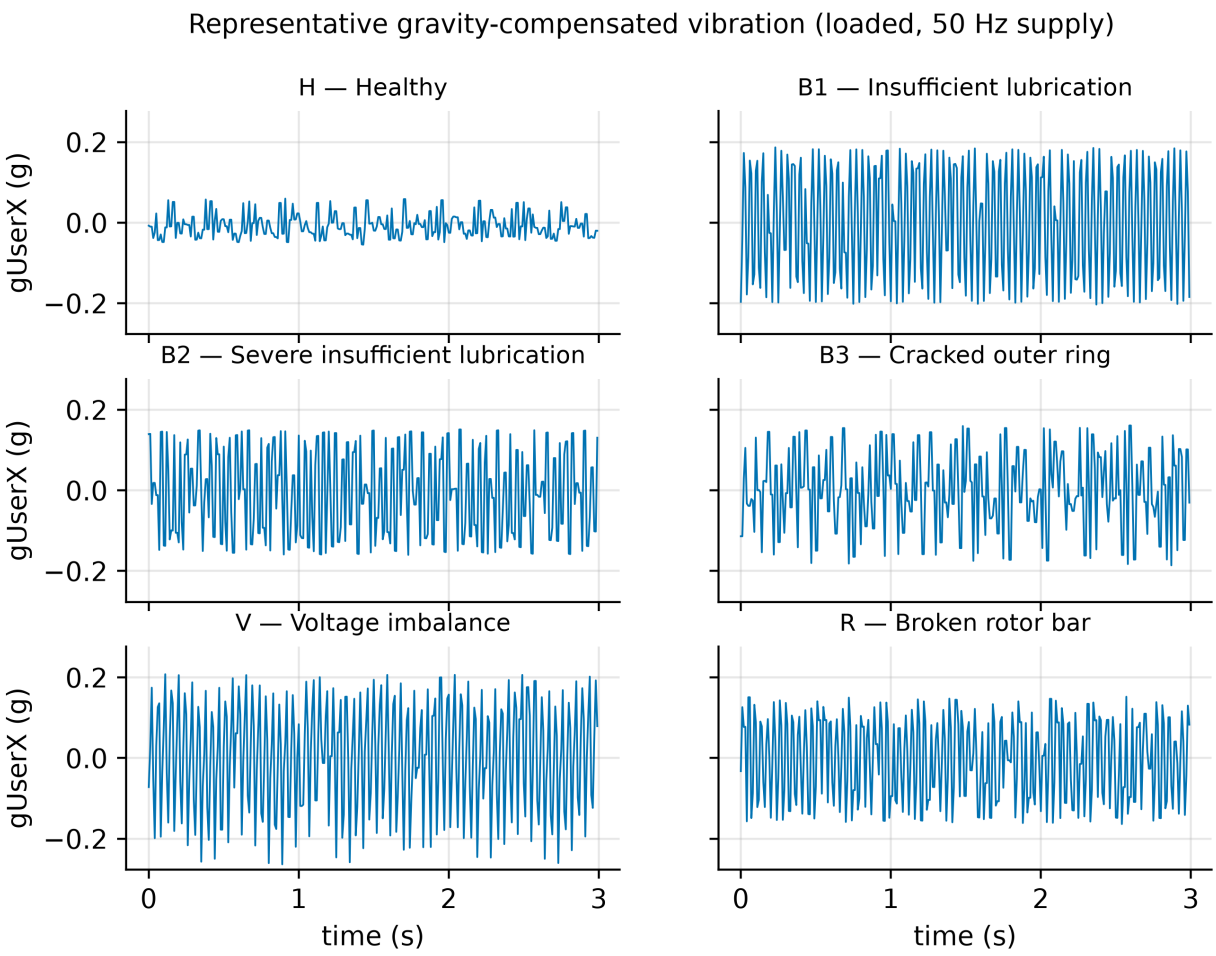

Figure 2: Representative gravity-compensated vibration (g_{userx}, 3 s excerpt) for the six health states under identical operating conditions (loaded, 50 Hz supply). Amplitudes share a common scale.

conflates the sensor with the device's internal processing, its acquisition software and the machine being measured. They are nonetheless the quantities that bound the diagnostic task, and they motivate the sensing choices analysed later.

4.1. Observed output increment and in-operation signal floor

The smallest positive difference between consecutive distinct sample values was extracted from a healthy, unloaded recording. This gives approximately 0.015 mg for the raw acceleration channels and 0.001 mg for the gravity-compensated channels. The quantity is the granularity of the delivered numerical stream, not the resolution of the transducer: the finer step of the gravity-compensated triad reflects the additional arithmetic of the on-board sensor fusion rather than a more sensitive sensing element, and neither figure can be interpreted as a noise-limited resolution without the reference measurements described above. An in-operation broadband floor was estimated from the PSD of a healthy recording; the median PSD above 30 Hz was 8.2×10^{-8} g^2/Hz (-70.8 dB re 1 g^2/Hz). Because this value is obtained on a *running* machine, it includes residual mechanical vibration and therefore upper-bounds, rather than isolates, the sensor's intrinsic noise floor, which would require an acquisition with the device at rest [40, 46]; no such recording exists in this dataset. Even as an upper bound, however, it lies several orders of magnitude below the fault-induced vibration, which reaches amplitudes on the order of 0.1–0.2 g (Fig. 2); sensor resolution and noise are thus not the limiting factor for the coarse fault-discrimination task considered here.

4.2. Band made observable by the sampling rate

At a sampling rate of 100 Hz the observable spectrum is bounded by the Nyquist frequency of 50 Hz [15]. This is a limit imposed by the acquisition settings, not a measured property of the accelerometer, whose own usable bandwidth is not accessible from these data. Fig. 3 shows the Welch PSD of the gravity-compensated g_{userx} channel for the six health states under identical operating conditions. Two observations follow. First, discriminative structure is present within the 0–50 Hz band: the healthy machine exhibits a spectrum roughly an order of magnitude lower in power than the fault conditions, and distinct spectral peaks appear for the rotor and electrical faults. The dominant peaks near 24 Hz coincide with the shaft rotational frequency at 50 Hz supply (Table 4, e.g. 1425–1486 r/min $\approx$ 23.8–24.8 Hz), confirming that the low-frequency band captures the rotational fundamental and its low-order harmonics. Second, the characteristic bearing defect frequencies (ball-pass and ball-spin frequencies and their harmonics) lie largely above 50 Hz at the higher supply frequencies—although some low-order components fall within the band at 30 Hz supply—and are, for the most part, not represented. The dataset is therefore better suited to time-domain and low-frequency analysis than to the high-frequency envelope analysis conventionally used for incipient bearing diagnosis, and a corresponding difficulty in separating the fine lubrication-severity states is anticipated.

4.3. Raw versus gravity-compensated channels

The dataset provides both the raw acceleration and the gravity-compensated (*userAcceleration*) triads, and their relative diagnostic value is not self-evident. Two questions were separated: whether the on-device gravity compensation removes vibration *power*, and whether it removes diagnostic *information*. For the first, the alternating-current (AC) power of each raw channel was compared with that of its gravity-compensated counterpart over every recording; the mean raw-to-compensated power ratio was 1.00 on all three axes (0.999, 1.001, 1.000 for x, y, and z). The compensation therefore does not attenuate the vibration power. It does, however, *reshape* it: the mean-removed raw signal and the compensated signal are only moderately correlated (Pearson 0.58, 0.46, and 0.46 for x, y, and z), so the complementary filter redistributes the low-frequency content—altering its phase and correlation structure—rather than discarding it, and the two triads are complementary rather than redundant.

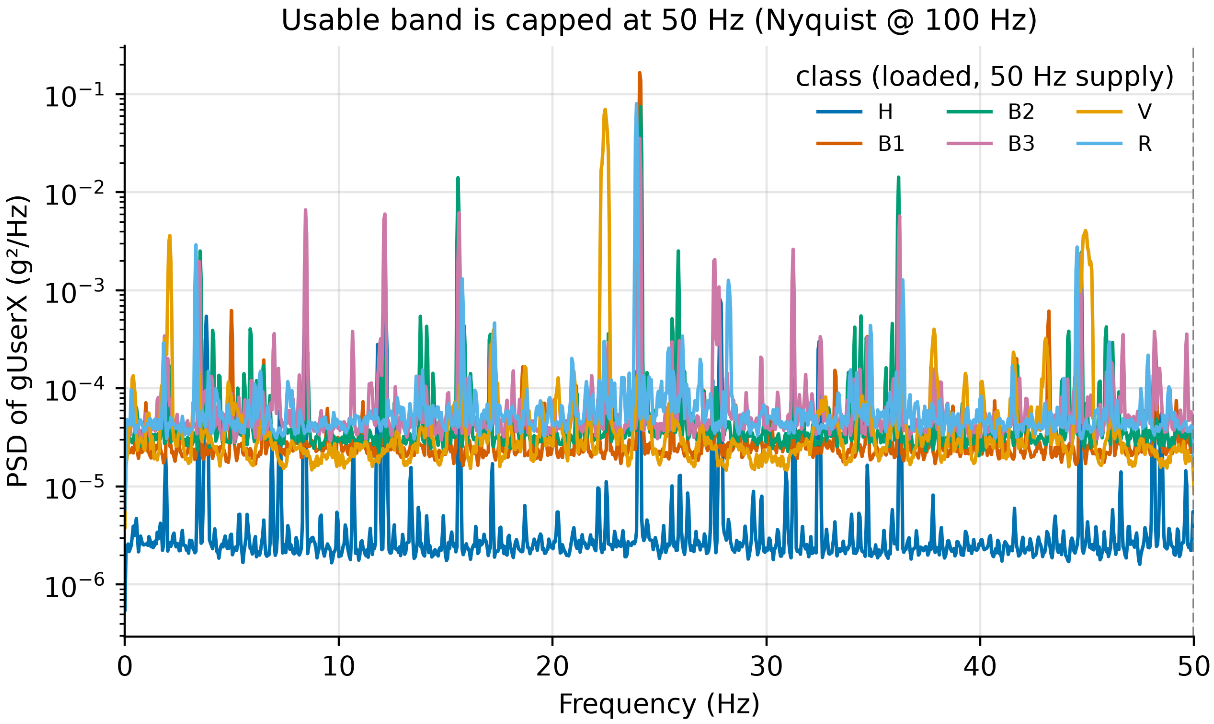

Figure 3: Welch power spectral density of the g_{userx} channel for the six health states (loaded, 50 Hz supply). The analysable band is bounded by the 50 Hz Nyquist frequency; the healthy state is markedly lower in power, and rotor/electrical faults produce distinct low-frequency peaks.

For the second question, the mutual information between each channel's time-domain feature set and the fault label was estimated (Fig. 4). The raw channels were consistently more informative (mean mutual information 0.69) than the gravity-compensated channels (0.50), a ratio of 1.39. Because the raw channels also retain a per-recording static offset that varies with condition, a legitimate concern is that this advantage merely encodes recording identity—the very leakage this work otherwise criticises. Two controls indicate that it does not: removing the offset-dependent *mean* feature leaves the ratio essentially unchanged (1.31), and restricting the estimate to a DC-invariant feature subset (standard deviation, variance, skewness, kurtosis, zero-crossing rate, and amplitude entropy) yields a comparable ratio (1.32). The raw-channel advantage is thus robust to the removal of static offsets and reflects genuine dynamic content. The raw channels are therefore preferable for condition monitoring, and this informativeness ordering motivates the channel-selection study reported in Section 6.

5. Methodology

The processing chain evaluated in this work is deliberately lightweight, so that it is representative of what an inexpensive sensor node can execute: fixed-length windowing, a compact set of time-domain features, and

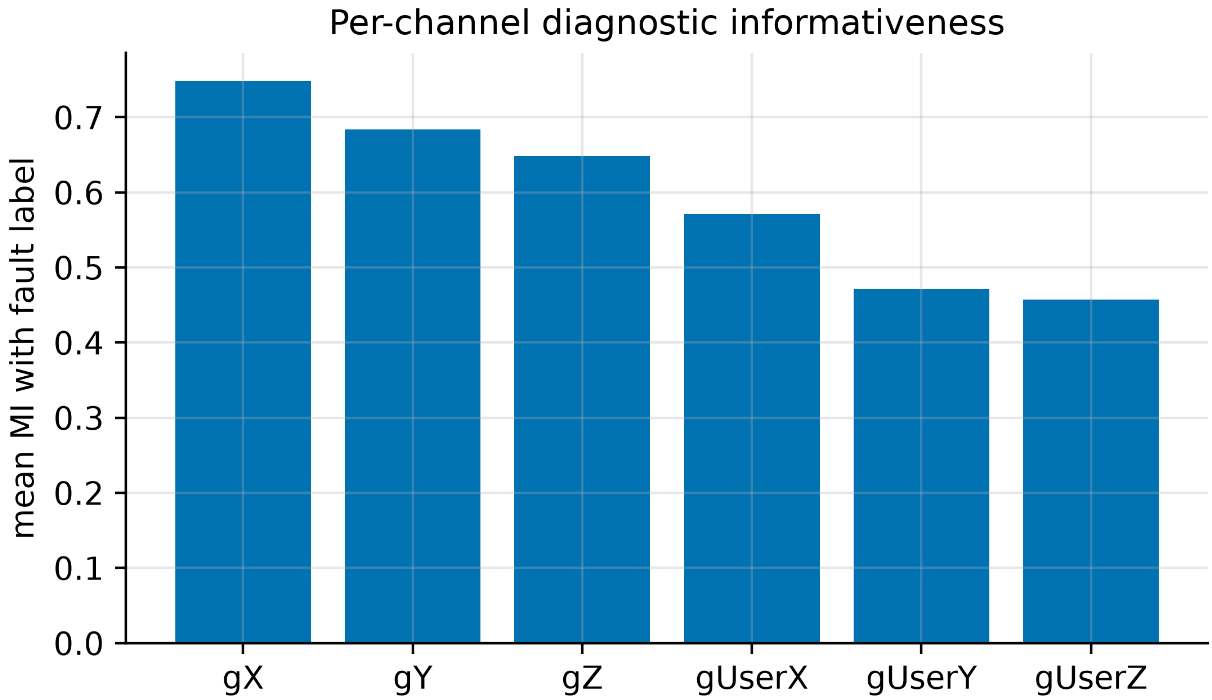

Figure 4: Mean mutual information between each channel's time-domain feature set and the fault label. The raw channels (g_x, g_y, g_z) are more informative than the gravity-compensated channels (g_{userx}, g_{usery}, g_{userz}).

a conventional classifier. The chain is illustrated in Fig. 5. The emphasis of this section, and the principal methodological contribution of the paper, is the set of *evaluation protocols* used to obtain deployment-realistic performance estimates.

5.1. Segmentation and Windowing

Let the dataset comprise $R = 36$ continuous recordings, each associated with a health label and an operating condition (supply frequency and load). Every recording was divided into fixed-length windows of 2 s with 50% overlap, matching the segmentation of the source data article. At 100 Hz this yields windows of 200 samples and a total of 32,328 windows, distributed equally across the six classes (5,388 per class). The window length and overlap were treated as configurable parameters and were varied in the trade-off study of Section 6.

5.2. Feature Extraction

Each window was reduced to a set of 14 time-domain features per channel, summarised in Table 6. These comprise standard statistical and shape descriptors widely used in vibration-based diagnosis, and require no spectral transform, which keeps the extraction cost compatible with a microcontroller. For a

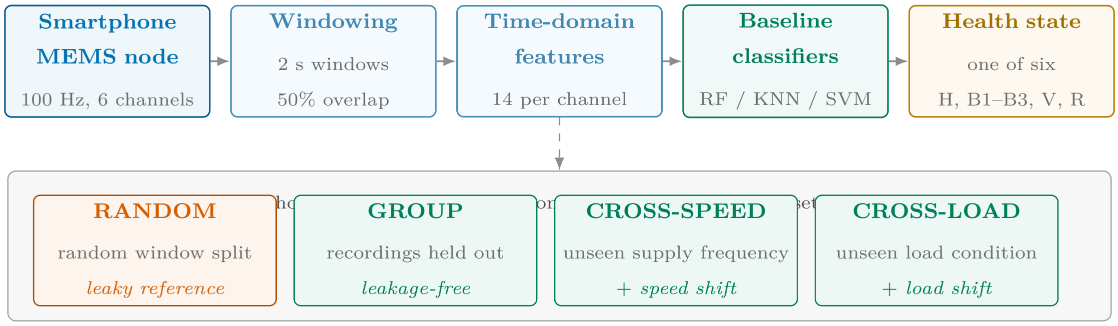

Figure 5: Processing chain and evaluation protocols. Identical features are assessed under four data-partitioning protocols: RANDOM is retained only as a leaky reference, whereas GROUP, CROSS-SPEED and CROSS-LOAD are leakage-free at the recording level, the latter two additionally imposing an operating-condition shift. The protocol, not the model, is the primary object of study.

window of C channels, the per-channel feature vectors were concatenated into a single descriptor of length $14C$ (for example, 84 for the full six-channel set). The six-feature subset marked in Table 6, corresponding to the descriptors used by the source data article's baseline, was retained for a feature-set ablation. For the distance- and margin-based classifiers, features were standardised to zero mean and unit variance using statistics estimated on the training partition only.

5.3. Classifiers

Three baseline classifiers were considered, chosen for their prevalence in the fault-diagnosis literature and their modest computational footprint: a Random Forest (RF) with 200 trees; a k-nearest-neighbour classifier (KNN, $k = 5$); and a support vector machine (SVM) with a radial-basis-function kernel. To keep the SVM tractable on the full window set, the kernel was approximated by a Nyström feature map (400 components) followed by a linear SVM. For the edge-footprint analysis of Section 6, two additional lightweight models—a depth-limited decision tree and a multinomial logistic regression—were included. All models map a feature descriptor to one of the six health states.

5.4. Evaluation Protocols

The central methodological question is how the labelled windows are partitioned into training and test sets. Four protocols were defined and applied to identical features so that differences in reported accuracy are attributable solely to the partitioning strategy.

5.4.1. RANDOM

A stratified random split over all windows, as commonly used in the literature and by the source baseline. This protocol is retained as a reference precisely because it is susceptible to leakage: adjacent windows share

50% of their samples, and all windows of a recording share the same operating point and sensor coupling, so a model may recognise the originating recording rather than the fault.

5.4.2. GROUP

A leakage-free protocol in which entire recordings are held out. Stratified group k-fold cross-validation ($k = 6$) was used, with the recording identifier as the grouping variable, so that no window from a test recording appears in training. With six recordings per class, each fold holds out one recording per class.

5.4.3. RANDOM without overlap and BLOCKED

Two further partitionings are defined to identify which property of the RANDOM protocol is responsible for its optimism. The first repeats the stratified random split on windows extracted with zero overlap, so that no sample can appear in both partitions. The second holds out a contiguous temporal block from within every recording, taking successive sixths of each recording in turn, so that training and test windows are additionally separated in time while every recording still contributes to training. Comparing these against RANDOM and GROUP attributes the gap to shared samples, to temporal proximity, or to recording identity.

5.4.4. CROSS-SPEED

A leave-one-supply-frequency-out protocol: the model is trained on two supply frequencies and tested on the third, and results are averaged over the three folds. This quantifies generalisation to an unseen operating speed.

5.4.5. CROSS-LOAD

A leave-one-load-out protocol: the model is trained on one load condition (loaded or unloaded) and tested on the other. This quantifies generalisation to an unseen load.

The GROUP, CROSS-SPEED, and CROSS-LOAD partitions are all group-disjoint at the recording level and therefore free of the window-overlap leakage that affects the RANDOM protocol; the latter two additionally introduce a deliberate operating -condition shift between training and test.

5.5. Reporting performance and its dispersion

Classification performance is reported as accuracy and macro-averaged F_1-score, the latter being sensitive to the per-class failures that a balanced accuracy can hide. Two questions then arise: over what unit is performance replicated, and how is that dispersion to be described.

The unit adopted here is the *held-out recording.* A score is computed separately for each recording while it is held out, so the reported mean is an average over 36 recordings and its dispersion is the recording-to-recording variation, which is the variation a new installation would actually encounter. Intervals are 95%

intervals for that mean, clustered on recording. Ten random seeds are used, each reshuffling the recording-level fold composition, but the seeds serve only to stabilise the per-recording means; they are not treated as independent observations, because re-partitioning the same 36 recordings supplies no new information about the machine and an interval computed over seeds understates the true dispersion by a large factor. In the present data the recording-clustered interval is roughly three times the seed-level interval, which is the size of the error that the more convenient choice would introduce.

For the same reason no claim of a Type A evaluation of measurement uncertainty in the sense of the GUM [65] is made for these figures, although that framework is the natural destination for such work [69, 66]. A GUM-conformant budget would require the dominant influence quantities—machine, fault specimen, mounting, session—to be varied and their contributions evaluated [42]; none of them varies within this dataset. What is reported is therefore a reproducibility interval over the available recordings, and the influence quantities that remain unquantified are set out qualitatively in Section 7.2.

Configurations are compared on their per-recording differences with a two-sided paired permutation test over the 2^n sign changes, approximated by 20 000 random sign assignments. The test is exact in construction, assumes neither normality nor equal variances, and—unlike a signed-rank test over five seeds—does not exhaust its resolution before reaching conventional significance levels. Confusion matrices are aggregated from the pooled out-of-fold predictions of the GROUP protocol.

Finally, where a configuration is selected rather than fixed in advance, selection and estimation are separated. Choosing an operating point by inspecting cross-validation scores and then quoting those same scores is optimistically biased [59]; Section **??** therefore repeats the selection inside a nested recording-wise loop, so that the recordings used to estimate performance are never seen by the selector.

5.6. Implementation Details

All estimators were taken from a single, standard machine-learning library, with the settings collected in Table 5; any parameter not listed was left at its library default, and every stochastic component was seeded. Three further choices affect reproducibility and are therefore stated explicitly. First, decimation to a lower sampling rate was performed by polyphase resampling with the associated anti-alias FIR filter, so that the reduced-rate conditions of Section 6.2 combine a change of rate with the low-pass that necessarily accompanies it. Second, the mutual information of Section 4 was obtained with a k-nearest-neighbour estimator ($k = 3$) applied to the continuous features and reported in nats; the estimator is non-negative by construction, so values near zero indicate no detectable dependence rather than independence. Third, the power spectral densities were computed by Welch's method with a Hann window, 50% segment overlap and density scaling, using 2048-sample segments for the per-class spectra of Fig. 3 and 1024-sample segments for the signal-floor estimate.

6. Results

6.1. The Optimism Gap of Random-Split Evaluation

Table 7 reports the classification accuracy of the three baseline models under the four evaluation protocols, and Fig. 6 visualises the same result. Under the RANDOM protocol every model attains essentially perfect accuracy, reproducing—and slightly exceeding—the 98% reported by the source data article's random-split baseline. Under the leakage-free GROUP protocol the accuracy of the same features and models falls to roughly one-half, an *optimism gap* of approximately 41 to 47 accuracy points. The CROSS-SPEED and CROSS-LOAD protocols, which additionally impose an operating-condition shift, yield comparably reduced accuracy.

6.1.1. Which mechanism produces the gap

A gap of that size is usually attributed to two causes at once: overlapping windows sharing samples across the partition boundary, and windows inheriting the identity of the recording they were cut from. These are separable, and a third candidate—simple temporal proximity between neighbouring windows—sits between them. Table 8 removes them one at a time. Discarding the 50% overlap halves the window count and guarantees that no sample appears in both partitions; holding out a contiguous temporal block within every recording additionally removes near-neighbour proximity while still training on every recording; and the recording-wise protocol finally removes recording identity as well.

The attribution is unambiguous. Removing window overlap changes nothing: a random split of strictly non-overlapping windows still classifies at 100.0%. Removing temporal proximity as well changes nothing either, with the blocked protocol also at 100.0%. The entire gap—46.7 points—appears at the last step, when whole recordings are held out. Sharing samples between partitions is therefore not the operative mechanism, contrary to the explanation usually offered. Every window cut from one recording carries that recording's particular mounting, operating point and session conditions, and this signature is apparently sufficient to identify the recording, and hence its label, from any window of it. Halving the windows or separating them in time leaves that signature intact. Only holding out the recording removes it.

This has a direct methodological consequence. Guarding against leakage by enforcing non-overlapping windows, a common precaution, would have offered no protection whatever on this dataset, and the same is likely wherever each operating condition is captured in a single continuous acquisition. The unit that must be held out is the recording.

Two further observations follow from Table 7: the leakage-free accuracy remains well above the 16.7% chance level, and the lower-capacity models (KNN, SVM) attain higher GROUP accuracy than the Random Forest.

The seed-averaged confusion matrix of the GROUP protocol (Fig. 7) localises the residual difficulty. The healthy state is the most reliably identified, followed by the supply-imbalance and broken-rotor-bar faults, whereas the two lubrication-severity states (B1 and B2) are frequently confused with each other and with the remaining bearing conditions, in agreement with the 50 Hz band limit established in Section 4. Because this per-class behaviour is markedly uneven, accuracy alone overstates how well the fine bearing states are resolved: the macro-averaged F_1-score of the same configuration is 51.1%, and per-class recall falls from 0.87 for the healthy state and 0.66 for the supply imbalance to 0.35 for B3 and 0.16 for B2. The lubrication-severity states therefore dominate the residual error, and the headline figures below should be read with that qualification.

6.2. Accuracy–Resource Trade-offs

Each sensing and communication parameter was varied in turn under the GROUP protocol (Fig. 8). Reducing the sampling rate did not degrade accuracy: as shown in Table 9, every rate at or below 25 Hz matched or exceeded the accuracy obtained at 100 Hz, with a maximum near 25 Hz. Reducing the channel set was likewise not detrimental—the three raw axes alone outperformed the full six-channel set—while lengthening the window from 2 s to 4 s increased accuracy at the cost of detection latency.

The combination of these frugal choices was then examined (Table 10 and Fig. 9). The resource-optimal configuration—25 Hz sampling, the three raw channels, and a 4 s window—raised the Random Forest accuracy from 52.8% at the naive 100 Hz/six-channel/2 s baseline to 75.9%, while reducing the raw data volume by approximately eight-fold. Robustness to further deployment stresses was also measured: reducing the amplitude resolution to 4 bits lowered accuracy by only a few points, injected packet loss of up to 20% left accuracy essentially unchanged, and the six-feature subset of the source baseline performed within noise of the full feature set. The largest sensitivity was to sensor mounting: a simulated 30° change in orientation degraded accuracy substantially.

The dispersion of these figures deserves care. Estimates were repeated over ten random recording-level partitions and scored separately on each held-out recording, so that the reported interval reflects recording-to-recording variation rather than the effect of re-partitioning the same data (Section 5.5). Table 11 gives the result. The intervals are wide—roughly ±12 points—because performance varies substantially from one recording to the next, and an interval computed over seeds instead would have been about three times narrower and correspondingly misleading.

Compared on their per-recording differences, the resource-frugal configuration classifies the health state 21.1 percentage points more accurately than the baseline, and a two-sided paired permutation test over the 36 recordings gives $p = 0.020$. The intermediate step tells a different story: reducing the sampling rate from 100 to 25 Hz alone accounts for 9.4 points, but that difference is not significant at the recording level ($p = 0.26$). The rate reduction on its own is therefore not established as an improvement by these data, even though the combination of rate reduction, channel selection and the longer window is. This distinction is invisible to a test conducted over repeated re-partitionings, which would have reported both differences as highly significant.

6.3. What the sampling-rate effect actually is

The sweep of Table 9 shows accuracy rising as the sampling rate falls, which invites the interpretation that the diagnostic information is concentrated at low frequency and that discarding the rest helps. That interpretation cannot be correct as stated. Shaft rotational frequencies in this dataset lie between 13.1 and 24.9 Hz (Table 4), so decimation to 25 Hz, whose Nyquist frequency is 12.5 Hz, removes every rotational fundamental in the dataset—including the peaks identified in Section 4—and accuracy nevertheless improves.

Three effects are confounded in a naive rate sweep: the band that survives, the number of samples the features are computed from, and the aliasing that occurs if the anti-alias filter is omitted. Table 12 separates them by holding each constant in turn. Band-limiting alone is obtained by low-pass filtering at 12.5 Hz while retaining the 100 Hz rate, so that the frequency content matches the decimated condition but the window still contains 200 samples; proper decimation to 25 Hz then reduces the window to 50 samples at the same bandwidth; and repeating the decimation with the anti-alias filter disabled isolates aliasing.

The outcome contradicts the simple reading. Restricting the band to 12.5 Hz at an unchanged sampling rate *reduces* accuracy by 2.7 points, so the content above 12.5 Hz is not merely useless—the full-rate configuration is using it. The entire improvement, 12.8 points, appears at the second step, where the only change is that the same band-limited signal is represented by a quarter as many samples per window. Disabling the anti-alias filter costs 24.9 points, which both confirms that the decimation is performed correctly and rules out any suggestion that aliasing folds useful content into the observable band.

The mechanism is therefore an interaction between the sampling rate and the feature set, not a property of the vibration signal alone. The 14 time-domain descriptors are not invariant to the sampling rate: zero-crossing rate, amplitude entropy over a fixed 32-bin histogram, and the higher-order moments are all computed over W samples, and at 100 Hz a signal band-limited to 12.5 Hz is four times oversampled, so consecutive samples are strongly correlated and the descriptors are estimated from far fewer effectively independent values than their sample count suggests. Decimation removes that redundancy. The practical consequence for a node is unchanged—25 Hz remains the better operating point—but the reason is not that the fault information lies below 12.5 Hz, and any study that sweeps sampling rate with a fixed time-domain feature set measures these two effects together rather than the effect of rate alone.

6.4. Deployment Implications: Communication and Edge Footprint

Table 13 compares the bitrate of three telemetry strategies with the sustainable throughput of common low-power radios. Streaming the raw six-channel signal requires 9.6 kbps and transmitting the extracted feature vector requires 2.69 kbps; both are carried by Bluetooth Low Energy, Zigbee, or NB-IoT, but neither fits within the duty-cycle-limited throughput of a LoRaWAN uplink, which admits only the 40 bps on-node

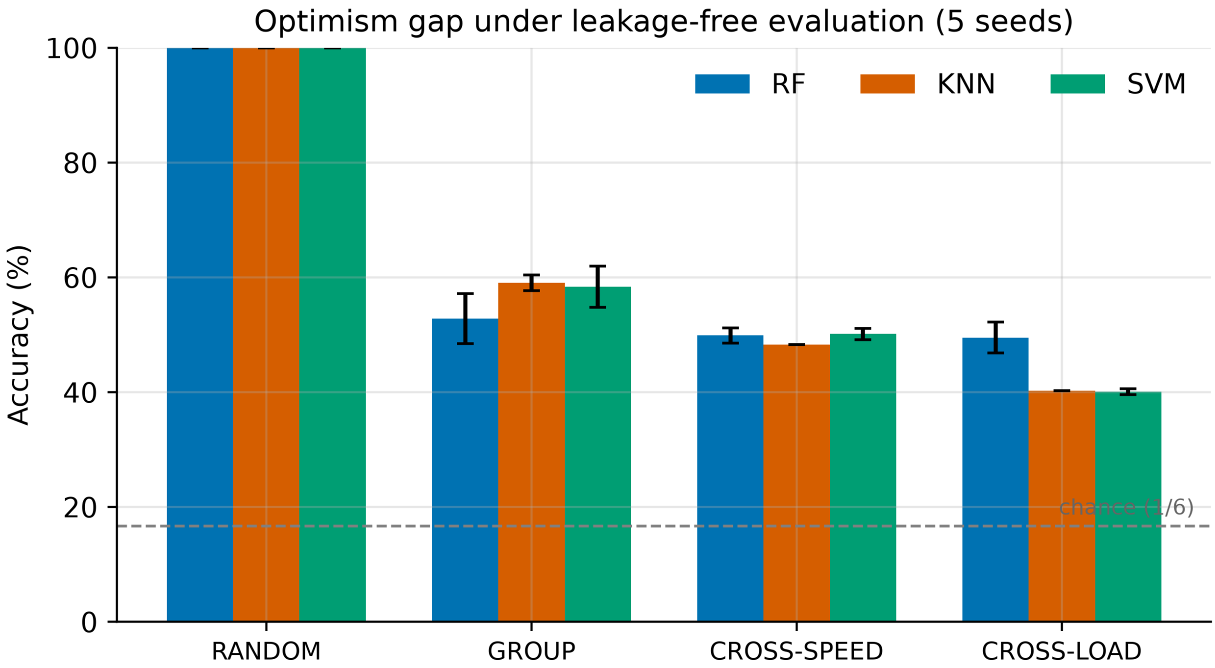

Figure 6: Accuracy by evaluation protocol and model. Near-perfect accuracy under the RANDOM split collapses once evaluation is leakage-free (GROUP) or imposes an operating-condition shift (CROSS-SPEED, CROSS-LOAD).

decision stream. Table 14 lists the leakage-free accuracy and serialised size of four candidate models: a 4.7 kB logistic-regression classifier exceeded the accuracy of a 1.09 MB Random Forest, and the time-domain feature extraction requires only $\mathcal{O}(WC)$ operations per window with no spectral transform.

7. Discussion and Limitations

7.1. Discussion

7.1.1. Evaluation methodology

The optimism gap of Section 6.1 is the central methodological finding: for low-rate smartphone-based machine diagnosis, accuracies obtained under random splitting overstate deployable performance by tens of accuracy points. The effect mirrors that documented for industrial-grade vibration data [18, 19] and the broader machine-learning leakage literature [17, 56], and it has been observed independently in wearable sensing [58].

The decomposition of Section 6.1.1 sharpens the usual account. Window overlap, which is the mechanism most often named and most often guarded against, accounts for none of the gap here: a random split of strictly non-overlapping windows is still perfectly accurate, as is a split that holds out a contiguous temporal block from every recording. The whole of the 46.7-point gap is attributable to recording identity. The practical implication is uncomfortable for common practice, because enforcing non-overlapping windows—a frequent and reasonable-looking precaution—would provide no protection at all on data of this kind. Where each operating condition is captured in one continuous acquisition, the recording is the only defensible unit of separation.

A consequence for benchmark practice follows. Near-perfect accuracies frequently reported for this class of data should be read as measurements of recording identifiability rather than of diagnostic capability, a concern raised independently for deep bearing-diagnosis benchmarks [60]. The recording-wise and cross-condition figures obtained here provide a more deployment-honest reference against which future methods can be compared.

7.1.2. Sampling rate, feature invariance and what the sweep measures

Decimation to 25 Hz raises leakage-free accuracy, and the resource-frugal configuration built on it remains the better operating point for a node. The reason, however, is not the one a rate sweep invites. Section 6.3 shows that band-limiting to the same 12.5 Hz while retaining the original sampling rate *reduces* accuracy, that the whole improvement appears when the band-limited signal is represented by a quarter as many samples per window, and that disabling the anti-alias filter is severely damaging. The shaft rotational frequencies of this machine, 13.1 to 24.9 Hz, lie above the 12.5 Hz Nyquist frequency of the decimated stream and are therefore absent from the configuration that performs best.

The operative effect is that the time-domain feature set is not invariant to the sampling rate. Zero-crossing rate, amplitude entropy over a fixed histogram and the higher-order moments all depend on the number of samples in the window and on how strongly those samples are correlated; a band-limited signal retained at 100 Hz is heavily oversampled, and its descriptors are estimated from far fewer effectively independent values than the sample count implies. Decimation removes that redundancy, and the gain follows.

Two conclusions are worth separating. The practical one is unchanged and useful: an ultra-low-rate node is not handicapped for this task, and aggressive decimation reduces telemetry and node energy without sacrificing accuracy—provided the anti-alias filter is applied, since omitting it costs 24.9 points. The methodological one is a caution that applies well beyond this dataset. A sampling rate sweep performed with a fixed time-domain feature set does not measure the information content of the band; it measures the band and the sample-count sensitivity of the features together. Reports that lower rates suffice, this one included, should be read with that confound in mind, and studies of downsampling in vibration analysis [50] suggest the interaction deserves explicit treatment.

7.1.3. Model complexity and edge feasibility

Under leakage-free evaluation the lower-capacity models generalised at least as well as the Random Forest while occupying a kilobyte-scale footprint, so on-node inference is practical on inexpensive microcontrollers. Combined with the link budget, this yields per-link deployment guidance: Bluetooth Low Energy, Zigbee, and NB-IoT nodes can stream raw or feature data to a gateway, whereas LoRaWAN nodes, constrained by the duty cycle to the decision stream, must perform classification locally. The alignment between the models that deploy most easily and those that generalise best is convenient for practical systems.

7.1.4. Sensor-level guidance

The channel analysis of Section 4 carries a concrete recommendation for smartphone-based acquisition: the raw acceleration channels should be preferred over the gravity-compensated channels. The on-device gravity compensation—valuable for motion-tracking applications—preserves the vibration power but reshapes its low-frequency structure, and the raw channels carry measurably more fault information (a mutual-information ratio of about 1.4 that survives the removal of static offsets), so they are the better choice for condition monitoring.

7.1.5. Scope of applicability

Taken together, the results delineate where low-cost, low-rate MEMS sensing is appropriate. Coarse health states, such as a healthy machine, a broken rotor bar, or a supply-voltage imbalance, are separable at 100 Hz and below; the staging of incipient bearing degradation, whose signatures lie in the high-frequency band, is not, and remains the province of higher-rate, industrial-grade instrumentation.

7.2. Limitations

Several limitations bound the scope of these findings. The dataset comprises a single 1.1 kW machine with laboratory-induced, discrete-severity faults, and the two lubrication states were graded qualitatively; the reported accuracies are therefore a deployment-honest baseline rather than a claim of field performance. Crucially, each fault class is represented by a *single physical specimen* (one modified bearing, one drilled rotor, one imbalance configuration) recorded across speeds and loads. The recording-wise protocol therefore removes window-overlap and exact operating-point leakage, but not specimen identity: a model may still key on the idiosyncratic signature of that particular specimen and its mounting rather than on a fault signature that would transfer to a different specimen of the same class. The evaluation is thus leakage-free only in this restricted sense, and cross-specimen generalisation is not established here. The small number of recordings (36) also produces wide fold-to-fold variability; although five-seed repetition stabilises the central estimates and the reported expanded uncertainty quantifies their reproducibility, it captures only resampling variance and not the dominant specimen-, machine-, and mounting-to-mounting components, so it should not be read as a full measurement-uncertainty budget. The deployment quantities are likewise indicative rather than measured: the model footprint is a serialised (software) size, and energy and latency are inferred from data volume and window length rather than measured on hardware, with a cycle-accurate microcontroller characterisation left to future work. A further limitation concerns reproducibility in practice: the implementation itself is not distributed, so the results must be reimplemented from the description rather than re-executed. The feature definitions, estimator settings, resampling and robustness transforms, and random seeds are therefore reported in full in Section 5, but exact numerical agreement with an independent reimplementation cannot be guaranteed. Finally, generalisation across machines of different ratings and manufacturers was not assessed and constitutes the most important direction for future validation.

8. Conclusion

This paper asked how the diagnostic performance of a low-cost, low-rate smartphone-MEMS sensing chain should be measured, and what that measurement then says about deploying such a chain on an induction machine. Using a public 100 Hz dataset, the near-perfect accuracies reported under random-split evaluation were shown to be an artefact of the evaluation protocol: under a recording-wise protocol the accuracy of standard classifiers falls by approximately 41 to 47 percentage points, with comparable reductions under cross-speed and cross-load shifts.

The more useful result is the attribution of that gap. Removing window overlap changes nothing, and removing temporal proximity between windows changes nothing either; the entire 46.7-point collapse occurs only when whole recordings are held out. Sharing samples across the partition boundary is therefore not the operative mechanism, and the precaution most commonly taken against leakage in this literature—non-overlapping windows—would have provided no protection whatever. Where each operating condition is captured in one continuous acquisition, the recording is the unit that must be separated.

A second finding concerns what a sampling-rate sweep measures. Decimation to 25 Hz improves leakage-free accuracy even though it removes every shaft rotational frequency present in the dataset. Band-limiting to the same range at the original rate is in fact mildly harmful, the improvement follows from the smaller number of samples per analysis window, and omitting the anti-alias filter costs 24.9 points. The time-domain feature set is not invariant to the sampling rate, so a rate sweep conducted with a fixed feature set measures the observable band and the sample-count sensitivity of the features at once. The practical guidance survives—a resource-frugal configuration of 25 Hz, three raw channels and a 4 s window reaches $75.0 \pm 11.4\%$ against $53.9 \pm 12.5\%$ for the full-rate baseline, a paired improvement of 21.1 points ($p = 0.020$), while cutting data volume roughly eight-fold—but the explanation for it changes, and the intermediate step of rate reduction alone is not significant at the recording level.

Reporting practice matters as much as the protocol. Scores were computed per held-out recording and summarised with intervals clustered on that unit, giving intervals of roughly ± 12 points; the same data summarised over repeated re-partitionings would have produced intervals about three times narrower and significance far stronger than the evidence supports. For a single-machine study in which each fault class is one physical specimen, that is the honest width.

Future work should extend the evaluation across machines of different ratings and manufacturers, and across repeated installations of the same fault, since mounting is known to be a first-order influence quantity and is the confound this design cannot address. Complementing the analysis with direct measurement of on-node inference latency and energy would replace the calculated deployment quantities with measured ones. The evaluation protocol set out here is intended to serve as a deployment-honest baseline for those efforts.

CRediT authorship contribution statement

Hüseyin Tayyer Canseven: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies

No generative AI tools were used to produce the scientific content, analysis or conclusions of this work.

Data availability

The data that support the findings of this study are openly available in Mendeley Data at `https://doi.org/10.17632/rs4vz8n3t5.1`, under a CC BY 4.0 licence, and are described in the accompanying data article [20, 24]. No new data were created in this study.

References

[1] P. Gangsar, R. Tiwari, Signal based condition monitoring techniques for fault detection and diagnosis of induction motors: A state-of-the-art review, Mechanical Systems and Signal Processing 144 (2020) 106908. doi:10.1016/j.ymssp.2020.106908. URL `http://dx.doi.org/10.1016/j.ymssp.2020.106908`

[2] K. Hamani, M. Kuchar, M. Kubatko, S. Kirschner, Advancements in induction motor fault diagnosis and condition monitoring: A comprehensive review, Sensors 25 (19) (2025) 5942. doi:10.3390/s25195942. URL `http://dx.doi.org/10.3390/s25195942`

[3] Y. Yang, M. M. M. Haque, D. Bai, W. Tang, Fault diagnosis of electric motors using deep learning algorithms and its application: A review, Energies 14 (21) (2021) 7017. doi:10.3390/en14217017.
URL http://dx.doi.org/10.3390/en14217017

[4] S. Zhang, S. Zhang, B. Wang, T. G. Habetler, Deep learning algorithms for bearing fault diagnostics—a comprehensive review, IEEE Access 8 (2020) 29857–29881. doi:10.1109/access.2020.2972859.
URL http://dx.doi.org/10.1109/ACCESS.2020.2972859

[5] S. Lee, T. Kim, T. Kim, Multi-domain vibration dataset with various bearing types under compound machine fault scenarios, Data in Brief 57 (2024) 110940. doi:10.1016/j.dib.2024.110940.
URL http://dx.doi.org/10.1016/j.dib.2024.110940

[6] M. Sehri, P. Dumond, University of ottawa constant and variable speed electric motor vibration and acoustic fault signature dataset, Data in Brief 53 (2024) 110144. doi:10.1016/j.dib.2024.110144.
URL http://dx.doi.org/10.1016/j.dib.2024.110144

[7] N. D. Thuan, H. S. Hong, Hust bearing: a practical dataset for ball bearing fault diagnosis, BMC Research Notes 16 (1) (2023). doi:10.1186/s13104-023-06400-4.
URL http://dx.doi.org/10.1186/s13104-023-06400-4

[8] H. Huang, N. Baddour, Bearing vibration data collected under time-varying rotational speed conditions, Data in Brief 21 (2018) 1745–1749. doi:10.1016/j.dib.2018.11.019.
URL http://dx.doi.org/10.1016/j.dib.2018.11.019

[9] G. Grouios, E. Ziagkas, A. Loukovitis, K. Chatzinikolaou, E. Koidou, Accelerometers in our pocket: Does smartphone accelerometer technology provide accurate data?, Sensors 23 (1) (2022) 192. doi:10.3390/s23010192.
URL http://dx.doi.org/10.3390/s23010192

[10] A. Naha, K. R. Thammayyabbabu, A. K. Samanta, A. Routray, A. K. Deb, Mobile application to detect induction motor faults, IEEE Embedded Systems Letters 9 (4) (2017) 117–120. doi:10.1109/les.2017.2734798.
URL http://dx.doi.org/10.1109/LES.2017.2734798

[11] M. Ertarğın, T. Gürgenç, Ö. Yıldırım, A. Orhan, A deep learning approach for motor fault detection using mobile accelerometer data, European Journal of Technic (Oct. 2023). doi:10.36222/ejt.1336342.
URL http://dx.doi.org/10.36222/ejt.1336342

[12] M. Ertarğın, A. Orhan, Ö. Yıldırım, T. Gürgenç, Automated fault classification of asynchronous motor using mobile phone accelerometer and parallel residual cnn-gru, Measurement 253 (2025) 117539. doi:10.1016/j.measurement.2025.117539.
URL http://dx.doi.org/10.1016/j.measurement.2025.117539

[13] L. A. dos Santos Pedotti, R. M. Zago, M. Giesbrecht, F. Fruett, Low-cost mems accelerometer network for rotating machine vibration diagnostics, IEEE Instrumentation & Measurement Magazine 23 (7) (2020) 25–33. doi:10.1109/mim.2020.9234762.
URL http://dx.doi.org/10.1109/mim.2020.9234762

[14] M. O. Jakobsen, Low cost mems accelerometer and microphone based condition monitoring sensor, with lora and bluetooth low energy radio, HardwareX 18 (2024) e00525. doi:10.1016/j.ohx.2024.e00525.
URL http://dx.doi.org/10.1016/j.ohx.2024.e00525

[15] H. Nyquist, Certain factors affecting telegraph speed, Transactions of the American Institute of Electrical Engineers XLIII (1924) 412–422. doi:10.1109/t-aiee.1924.5060996.
URL http://dx.doi.org/10.1109/T-AIEE.1924.5060996

[16] C. E. Shannon, A mathematical theory of communication, Bell System Technical Journal 27 (3) (1948) 379–423. doi:10.1002/j.1538-7305.1948.tb01338.x.
URL http://dx.doi.org/10.1002/j.1538-7305.1948.tb01338.x

[17] S. Kapoor, A. Narayanan, Leakage and the reproducibility crisis in machine-learning-based science, Patterns 4 (9) (2023) 100804. doi:10.1016/j.patter.2023.100804.
URL http://dx.doi.org/10.1016/j.patter.2023.100804

[18] L. Wheat, M. V. Mohrenschildt, S. Habibi, D. Al-Ani, Impact of data leakage in vibration signals used for bearing fault diagnosis, IEEE Access 12 (2024) 169879–169895. doi:10.1109/access.2024.3497716.
URL http://dx.doi.org/10.1109/ACCESS.2024.3497716

[19] J. Hendriks, P. Dumond, D. Knox, Towards better benchmarking using the cwru bearing fault dataset, Mechanical Systems and Signal Processing 169 (2022) 108732. doi:10.1016/j.ymssp.2021.108732.
URL http://dx.doi.org/10.1016/j.ymssp.2021.108732

[20] M. Ertarğın, A. Orhan, Ö. Yıldırım, S. Chauhan, G. Vashishtha, A smartphone-based vibration dataset for induction motor fault diagnosis under different speed and load conditions, Data in Brief 67 (2026) 112916. doi:10.1016/j.dib.2026.112916.
URL http://dx.doi.org/10.1016/j.dib.2026.112916

[21] M. Yousuf, T. Alsuwian, A. A. Amin, S. Fareed, M. Hamza, Iot-based health monitoring and fault detection of industrial ac induction motor for efficient predictive maintenance, Measurement and Control 57 (8) (2024) 1146–1160. doi:10.1177/00202940241231473.
URL http://dx.doi.org/10.1177/00202940241231473

[22] Z. Gao, Z. Jiang, Z. Dong, X. Fu, Y. Chen, C. Zhang, An edge-deployable tinyml approach enhanced by transfer learning for efficient bearing fault diagnosis, Science China Technological Sciences 68 (12) (Nov. 2025). doi:10.1007/s11431-025-3072-9.
URL http://dx.doi.org/10.1007/s11431-025-3072-9

[23] S. Asutkar, C. Chalke, K. Shivgan, S. Tallur, Tinyml-enabled edge implementation of transfer learning framework for domain generalization in machine fault diagnosis, Expert Systems with Applications 213 (2023) 119016. doi:10.1016/j.eswa.2022.119016.
URL http://dx.doi.org/10.1016/j.eswa.2022.119016

[24] M. Ertarğın, A. Orhan, Ö. Yıldırım, S. Chauhan, G. Vashishtha, A smartphone-based vibration dataset for induction motor fault diagnosis under different speed and load conditions (2026). doi:10.17632/RS4VZ8N3T5.1.
URL https://data.mendeley.com/datasets/rs4vz8n3t5/1

[25] Y. Liu, Y. Cheng, S. Yang, J. Wu, Vibration signal models for fault diagnosis of rolling element bearing, Measurement 264 (2026) 120303. doi:10.1016/j.measurement.2026.120303.
URL http://dx.doi.org/10.1016/j.measurement.2026.120303

[26] J. Li, W. Luo, M. Bai, Review of research on signal decomposition and fault diagnosis of rolling bearing based on vibration signal, Measurement Science and Technology 35 (9) (2024) 092001. doi:10.1088/1361-6501/ad4eff.
URL http://dx.doi.org/10.1088/1361-6501/ad4eff

[27] S. Ayankoso, A. Dutta, Y. He, F. Gu, A. Ball, S. K. Pal, Performance of vibration and current signals in the fault diagnosis of induction motors using deep learning and machine learning techniques, Structural Health Monitoring 25 (1) (2024) 196–212. doi:10.1177/14759217241289874.
URL http://dx.doi.org/10.1177/14759217241289874

[28] Q. Sun, C. Chen, X. Liu, Comparative study of stator current-based and vibration-based methods for railway traction motor bearing cage fault diagnosis at high-speed condition, Structural Health Monitoring 22 (2) (2022) 978–992. doi:10.1177/14759217221104217.
URL http://dx.doi.org/10.1177/14759217221104217

[29] J. Guo, Q. He, F. Gu, Dnocnet: A novel end-to-end network for induction motor drive systems fault diagnosis under speed fluctuation condition, IEEE Transactions on Industrial Informatics 20 (6) (2024) 8284–8293. doi:10.1109/tii.2024.3369239.

URL http://dx.doi.org/10.1109/tii.2024.3369239

[30] X. Chen, Z. Chen, J. Xin, L. Guo, W. Zhai, Discriminative condition-guided generative model for induction motor fault diagnosis with limited data, IEEE Transactions on Industrial Informatics 22 (2) (2026) 1359–1370. doi:10.1109/tii.2025.3626126.
URL http://dx.doi.org/10.1109/tii.2025.3626126

[31] A. Prato, F. Mazzoleni, G. D'Emilia, A. Gaspari, E. Natale, A. Schiavi, Metrological traceability of a digital 3-axis mems accelerometers sensor network, Measurement 184 (2021) 109925. doi:10.1016/j.measurement.2021.109925.
URL http://dx.doi.org/10.1016/j.measurement.2021.109925

[32] A. Schiavi, A. Prato, F. Mazzoleni, A. Facello, A. Kwitonda, Metrological characterization of digital mems accelerometers in dynamic conditions: An investigation of linearity in amplitude and temperature effects, in: Proceedings of the 5th IMEKO TC22 Conference on Vibration Measurement, IMEKO, 2023, pp. 1–4. doi:10.21014/tc22-2022.033.
URL http://dx.doi.org/10.21014/tc22-2022.033

[33] D. Wang, F. Li, N. Zhang, L. Xu, F. Lyu, In-field mems accelerometer calibration considering cross-axis sensitivities without external equipment, IEEE Sensors Journal 25 (10) (2025) 16981–16990. doi:10.1109/jsen.2025.3557731.
URL http://dx.doi.org/10.1109/jsen.2025.3557731

[34] H. R. Najafabadi, T. G. Goto, T. C. Martins, M. S. Tsuzuki, A. Barari, Designing mems accelerometer for enhanced sensitivity and reduced cross-sensitivity in landslide monitoring, Measurement 226 (2024) 114092. doi:10.1016/j.measurement.2023.114092.
URL http://dx.doi.org/10.1016/j.measurement.2023.114092

[35] T. Xu, X. Xu, D. Xu, Z. Zou, H. Zhao, Low-cost and efficient thermal calibration scheme for mems triaxial accelerometer, IEEE Transactions on Instrumentation and Measurement 70 (2021) 1–9. doi:10.1109/tim.2021.3096290.
URL http://dx.doi.org/10.1109/tim.2021.3096290

[36] H. J. Kim, H. K. Jung, Temperature hysteresis calibration method of mems accelerometer, Sensors 25 (19) (2025) 6131. doi:10.3390/s25196131.
URL http://dx.doi.org/10.3390/s25196131

[37] S. Khankalantary, S. Ranjbaran, S. Ebadollahi, Simplification of calibration of low-cost mems accelerometer and its temperature compensation without accurate laboratory equipment, Measurement Science and Technology 32 (4) (2021) 045102. doi:10.1088/1361-6501/abd0bf.
URL http://dx.doi.org/10.1088/1361-6501/abd0bf

[38] Y. Li, L. Song, S. Liang, Y. Xiao, F. Yang, Nonlinear vibration study based on uncertainty analysis in mems resonant accelerometer, Sensors 20 (24) (2020) 7207. doi:10.3390/s20247207.
URL http://dx.doi.org/10.3390/s20247207

[39] T. Xu, X. Xu, D. Xu, H. Zhao, A novel calibration method using six positions for mems triaxial accelerometer, IEEE Transactions on Instrumentation and Measurement 70 (2021) 1–11. doi:10.1109/tim.2020.3026024.
URL http://dx.doi.org/10.1109/tim.2020.3026024

[40] P. E. Srokosz, E. Daniszewska, J. Banach, M. Smieja, In-depth analysis of low-cost micro electromechanical system (mems) accelerometers in the context of low frequencies and vibration amplitudes, Sensors 24 (21) (2024) 6877. doi:10.3390/s24216877.
URL http://dx.doi.org/10.3390/s24216877

[41] G. D'Emilia, E. Natale, Use of mems sensors for condition monitoring of devices: discussion about the accuracy of features for diagnosis, International Journal of Metrology and Quality Engineering 12 (2021) 15. doi:10.1051/ijmqe/2021011.
URL http://dx.doi.org/10.1051/ijmqe/2021011

[42] W. Staszewski, A. Jablonski, Analysis of impact of mems accelerometer positioning on vibration sensor performance,

Metrology and Measurement Systems (2025) 1–15doi:10.24425/mms.2025.155798.
URL http://dx.doi.org/10.24425/mms.2025.155798

[43] J. Lee, K. Kim, D. Kim, Z. Ma, H. Sohn, Improved structural acceleration estimation using low-cost mems accelerometer and fmcw millimeter-wave radar, Measurement 270 (2026) 120848. doi:10.1016/j.measurement.2026.120848.
URL http://dx.doi.org/10.1016/j.measurement.2026.120848

[44] A. Sayghe, Reliability-aware vibration measurement framework for predictive maintenance of rotating machinery: Probability of detection, uncertainty-quantified remaining useful life, and simulator-based hardware-in-the-loop reliability evaluation, Measurement 286 (2026) 122345. doi:10.1016/j.measurement.2026.122345.
URL http://dx.doi.org/10.1016/j.measurement.2026.122345

[45] A. Shukla, M. Mahmud, W. Wang, A smart sensor-based monitoring system for vibration measurement and bearing fault detection, Measurement Science and Technology 31 (10) (2020) 105104. doi:10.1088/1361-6501/ab8dfc.
URL http://dx.doi.org/10.1088/1361-6501/ab8dfc

[46] V. Capuano, L. Xu, J. Estrada Benavides, Smartphone mems accelerometer and gyroscope measurement errors: Laboratory testing and analysis of the effects on positioning performance, Sensors 23 (17) (2023) 7609. doi:10.3390/s23177609.
URL http://dx.doi.org/10.3390/s23177609

[47] B. Zhao, T. Nagayama, K. Xue, Road profile estimation, and its numerical and experimental validation, by smartphone measurement of the dynamic responses of an ordinary vehicle, Journal of Sound and Vibration 457 (2019) 92–117. doi:10.1016/j.jsv.2019.05.015.
URL http://dx.doi.org/10.1016/j.jsv.2019.05.015

[48] D. K. Periyasamy, M. V. Sivasubramanian, B. Surendiran, A time-synchronized multi-smartphone data acquisition system for structural vibration monitoring, Measurement 279 (2026) 121610. doi:10.1016/j.measurement.2026.121610.
URL http://dx.doi.org/10.1016/j.measurement.2026.121610

[49] L. Yang, X. Li, W. Wang, X. Wang, P. Li, G. Liu, Z. Wang, Submillimeter acoustic vibration measurement and monitoring using a single smartphone, IEEE Transactions on Instrumentation and Measurement 74 (2025) 1–15. doi:10.1109/tim.2025.3552861.
URL http://dx.doi.org/10.1109/tim.2025.3552861

[50] D. He, H. Cao, Downsampling-based synchrosqueezing transform and its applications on large-scale vibration data, Journal of Sound and Vibration 496 (2021) 115938. doi:10.1016/j.jsv.2021.115938.
URL http://dx.doi.org/10.1016/j.jsv.2021.115938

[51] I. Halenar, L. Halenarova, P. Tanuska, P. Vazan, Machine condition monitoring system based on edge computing technology, Sensors 25 (1) (2024) 180. doi:10.3390/s25010180.
URL http://dx.doi.org/10.3390/s25010180

[52] S. A. R. Zaidi, A. M. Hayajneh, M. Hafeez, Q. Z. Ahmed, Unlocking edge intelligence through tiny machine learning (tinyml), IEEE Access 10 (2022) 100867–100877. doi:10.1109/access.2022.3207200.
URL http://dx.doi.org/10.1109/access.2022.3207200

[53] J. Chaoraingern, A. Numsomran, Embedded sensor data fusion and tinyml for real-time remaining useful life estimation of uav li polymer batteries, Sensors 25 (12) (2025) 3810. doi:10.3390/s25123810.
URL http://dx.doi.org/10.3390/s25123810

[54] O. Rubes, J. Chalupa, F. Ksica, Z. Hadas, Development and experimental validation of self-powered wireless vibration sensor node using vibration energy harvester, Mechanical Systems and Signal Processing 160 (2021) 107890. doi:10.1016/j.ymssp.2021.107890.
URL http://dx.doi.org/10.1016/j.ymssp.2021.107890

[55] H. Qiao, H. Guan, A. Jabbour, Y. Zhu, Synchronous lora sensor nodes for modal identification in footbridge vibration

monitoring, Journal of Civil Structural Health Monitoring 15 (2) (2024) 697–716. doi:10.1007/s13349-024-00864-z.
URL http://dx.doi.org/10.1007/s13349-024-00864-z

[56] M. A. Lones, Avoiding common machine learning pitfalls, Patterns 5 (10) (2024) 101046. doi:10.1016/j.patter.2024.101046.
URL http://dx.doi.org/10.1016/j.patter.2024.101046

[57] E. Ursu, A. Minnegalieva, P. Rawat, M. Chernigovskaya, R. Tacutu, G. K. Sandve, P. A. Robert, V. Greiff, Training data composition determines machine learning generalization and biological rule discovery, Nature Machine Intelligence 7 (8) (2025) 1206–1219. doi:10.1038/s42256-025-01089-5.
URL http://dx.doi.org/10.1038/s42256-025-01089-5

[58] D. Gholamiangonabadi, N. Kiselov, K. Grolinger, Deep neural networks for human activity recognition with wearable sensors: Leave-one-subject-out cross-validation for model selection, IEEE Access 8 (2020) 133982–133994. doi:10.1109/access.2020.3010715.
URL http://dx.doi.org/10.1109/access.2020.3010715

[59] C. Qi, J. Diao, L. Qiu, On estimating model in feature selection with cross-validation, IEEE Access 7 (2019) 33454–33463. doi:10.1109/access.2019.2892062.
URL http://dx.doi.org/10.1109/access.2019.2892062

[60] Y. Wang, Y. Tang, Silent collapse to a dataset-dependent attractor class in deep bearing fault diagnosis: a cross-dataset study of failure modes masked by evaluation practice, Measurement Science and Technology 37 (33) (2026) 335009. doi:10.1088/1361-6501/ae9438.
URL http://dx.doi.org/10.1088/1361-6501/ae9438

[61] A.-S. Qin, H.-L. Mao, Q. Hu, Cross-domain fault diagnosis of rolling bearing using similar features-based transfer approach, Measurement 172 (2021) 108900. doi:10.1016/j.measurement.2020.108900.
URL http://dx.doi.org/10.1016/j.measurement.2020.108900

[62] Y. Zou, Y. Liu, J. Deng, Y. Jiang, W. Zhang, A novel transfer learning method for bearing fault diagnosis under different working conditions, Measurement 171 (2021) 108767. doi:10.1016/j.measurement.2020.108767.
URL http://dx.doi.org/10.1016/j.measurement.2020.108767

[63] K. Huang, Z. Ren, L. Zhu, T. Lin, Y. Zhu, L. Zeng, J. Wan, A three-stage bearing transfer fault diagnosis method for large domain shift scenarios, Reliability Engineering & System Safety 254 (2025) 110641. doi:10.1016/j.ress.2024.110641.
URL http://dx.doi.org/10.1016/j.ress.2024.110641

[64] Ş. M. Kaya, A. E. Jobani, Iot-driven robust bearing fault diagnosis for induction motors under operating-condition shift, Sensors 26 (12) (2026) 3829. doi:10.3390/s26123829.
URL http://dx.doi.org/10.3390/s26123829

[65] JCGM, Evaluation of measurement data—guide to the expression of uncertainty in measurement (GUM), JCGM 100:2008, joint Committee for Guides in Metrology (2008).

[66] M. Dorozhovets, Forward and inverse problems of type a uncertainty evaluation, Measurement 165 (2020) 108072. doi:10.1016/j.measurement.2020.108072.
URL http://dx.doi.org/10.1016/j.measurement.2020.108072

[67] M. Dorozhovets, Evaluation of the type a uncertainty caused by simultaneous influence of random noise and power line interference, Metrology and Measurement Systems (2025) 1–15doi:10.24425/mms.2025.154340.
URL http://dx.doi.org/10.24425/mms.2025.154340

[68] D. Ou, Y. Xiao, D. Lan, Y. Wang, Type a standard uncertainty evaluation in one measurement through uncertainty propagation from voxel values' distribution for computed tomography metrology, Measurement Science and Technology 35 (6) (2024) 065015. doi:10.1088/1361-6501/ad30ba.
URL http://dx.doi.org/10.1088/1361-6501/ad30ba

[69] L. Angrisani, P. Arpaia, S. Criscuolo, M. D'Arco, E. De Benedetto, L. Duraccio, A. Tedesco, Machine learning-based measurement models: A novel methodology for gum-based uncertainty evaluation, IEEE Transactions on Instrumentation and Measurement 74 (2025) 1–11. doi:10.1109/tim.2025.3643048.
URL `http://dx.doi.org/10.1109/tim.2025.3643048`

[70] R. Morello, Gum-based decisional criteria to make decisions in presence of measurement uncertainty, IEEE Transactions on Instrumentation and Measurement 69 (8) (2020) 5511–5522. doi:10.1109/tim.2019.2963581.
URL `http://dx.doi.org/10.1109/tim.2019.2963581`

Table 1: Positioning of this work against representative prior studies. ✓ = addressed, ∼ = partially addressed, ✗ = not addressed.

Representative work(s)	**Sensor / rate**	**Low cost**	**Leakage-free evaluation**	**Acc.–resource trade-off**	**IoT/edge analysis**	**Sensor charact.**
High-rate vibration FD & deep models [5, 6, 7, 4, 29]	Industrial accel., 8–200 kHz	✗	✗	✗	✗	✗
Evaluation / leakage studies [17, 18, 19, 56, 59]	Industrial accel., k Hz	✗	✓	✗	✗	✗
MEMS metrology [31, 32, 39, 33, 42]	MEMS, bench setup	✓	n/a	✗	✗	✓
Smartphone / mobile machine FD [10, 11, 12]	Smartphone MEMS, $\leq$ k Hz	✓	✗	✗	∼	✗
Source-dataset baseline [20]	Smartphone MEMS, 100 Hz	✓	✗	✗	✗	∼
Low-cost MEMS sensor nodes [13, 14, 54]	MEMS node, k Hz	✓	✗	∼	✓	∼
TinyML edge fault diagnosis [22, 23, 53]	(Lab) accel., k Hz	✓	✗	∼	✓	✗
This work	**Smartphone-grade MEMS, 100 Hz**	✓	✓	✓	∼	∼

Table 2: Acquisition and dataset specifications.

Parameter	**Value**
Machine	1.1 kW three-phase squirrel-cage IM
Speed control	V/f-controlled inverter
Loading	DC generator (shaft-coupled)
Sensor	iPhone 15 Pro Max, tri-axial MEMS
Acquisition app	Sensor Play
Channels	6 (g_x, g_y, g_z; $g_{userx}, g_{usery}, g_{userz}$)
Sampling rate	100 Hz
Record length	15 min (89,999 samples/channel)
Health states	6 (H, B1, B2, B3, V, R)
Supply frequencies	30, 40, 50 Hz
Load conditions	loaded, unloaded
Recordings	36 ($6 \times 3 \times 2$)
Formats	`.csv`, `.mat`
Repository	Mendeley Data, CC BY 4.0

Table 3: Health states and fault-emulation summary.

Code	Health state	Emulation
H	Healthy	Fault-free baseline
B1	Insufficient lubrication	Reduced lubricant, 6205ZZ bearing; severity set by manual inspection
B2	Severe insufficient lubrication	Advanced lubricant starvation; severity set by manual inspection
B3	Cracked outer ring	Bearing with cracked outer ring (ex-service degradation)
V	Voltage imbalance	Voltage-divider resistors in one supply phase; phase-current asymmetry
R	Broken rotor bar	Four holes drilled in rotor bars (5–9 mm diameter, 16–18 mm depth)

Table 4: Measured shaft speed (r/min) for each health state, supply frequency, and load condition (reproduced from [20]).

Supply	Load	H	B1	B2	B3	V	R
30 Hz	Loaded	833.9	829.4	833.0	833.8	787.0	815.8
	Unloaded	892.7	888.3	892.7	896.3	893.0	894.9
40 Hz	Loaded	1136	1138	1138	1140	1076	1122
	Unloaded	1190	1192	1193	1192	1192	1192
50 Hz	Loaded	1434	1436	1439	1436	1326	1425
	Unloaded	1486	1490	1491	1489	1486	1486

Table 5: Estimator and feature-extraction settings. Parameters not listed were left at their library defaults; d denotes the feature-vector dimension ($14C$ for C channels).

Component	**Setting**
Random Forest (baseline)	200 trees, unlimited depth, Gini split, $\sqrt{d}$ features per split
Random Forest (edge)	50 trees, maximum depth 8
Decision tree (edge)	maximum depth 8
KNN	$k = 5$, Euclidean metric, uniform weights
SVM	RBF kernel approximated by a Nyström map (400 components, $\gamma = 1/d$), then a linear SVM ($C = 10$, squared hinge, primal solver, ≤ 5000 iterations)
Logistic regression	multinomial, L_2 penalty, $C = 1$, L-BFGS, ≤ 2000 iterations
Standardisation	zero mean, unit variance; fitted on the training partition only (KNN, SVM, logistic regression)
Amplitude entropy	32-bin normalised histogram per window
Bit-depth reduction	per-recording, per-channel min–max range uniformly quantised to $2^b - 1$ levels
Packet loss	independent Bernoulli per sample, concealed by sample-and-hold
Orientation change	Rodrigues rotation of both triads by 30° about the axis $(1, 1, 0)/\sqrt{2}$

Table 6: Per-channel time-domain features. For a window $x = \{x_i\}_{i=1}^{W}$ with mean μ and standard deviation σ. Features marked † form the six-feature ablation subset.

Feature	**Definition**
Mean	$\mu = \frac{1}{W}\sum_i x_i$
Std. dev.†	$\sigma = \sqrt{\frac{1}{W}\sum_i (x_i - \mu)^2}$
Variance	σ^2
RMS†	$x_{\text{rms}} = \sqrt{\frac{1}{W}\sum_i x_i^2}$
Peak-to-peak†	$\max_i x_i - \min_i x_i$
Max. absolute	$x_{\max} = \max_i \lvert x_i \rvert$
Skewness†	$\frac{1}{W}\sum_i (x_i - \mu)^3 / \sigma^3$
Kurtosis†	$\frac{1}{W}\sum_i (x_i - \mu)^4 / \sigma^4 - 3$
Crest factor†	$x_{\max}/x_{\text{rms}}$
Shape factor	$x_{\text{rms}} / \big(\frac{1}{W}\sum_i \lvert x_i \rvert\big)$
Impulse factor	$x_{\max} / \big(\frac{1}{W}\sum_i \lvert x_i \rvert\big)$
Clearance factor	$x_{\max} / \big(\frac{1}{W}\sum_i \sqrt{\lvert x_i \rvert}\big)^2$
Zero-crossing rate	rate of sign changes of $x_i - \mu$
Amplitude entropy	$-\sum_k p_k \log_2 p_k$ (32-bin histogram)

Table 7: Accuracy (%) by evaluation protocol and model (five-seed means; 100 Hz, six channels, 2 s. Dispersion on the recording unit is reported in Tables eftab:decomp and eftab:measured).

Protocol	**RF**	**KNN**	**SVM**
RANDOM (leaky)	100.0	100.0	100.0
GROUP (recording-holdout)	52.8	59.0	58.4
CROSS-SPEED	49.9	48.2	50.1
CROSS-LOAD	49.5	40.3	40.1

Table 8: Attribution of the optimism gap (Random Forest, five seeds; intervals are 95% clustered on recording, $n = 36$). Each row removes one further source of information shared between the training and test partitions.

Partitioning	**Sources retained**	**Accuracy (%)**	**Macro-F_1 (%)**
RANDOM, 50% overlap	samples, proximity, identity	100.0	100.0
RANDOM, no overlap	proximity, identity	100.0	100.0
BLOCKED within recording	identity	100.0	100.0
GROUP, no overlap	none	53.3	49.1
GROUP, 50% overlap	none	52.8	48.4
CROSS-SPEED	none, + speed shift	49.9	41.5
CROSS-LOAD	none, + load shift	49.5	44.5
Attribution			
shared samples (window overlap)		+0.0	
temporal proximity		+0.0	
recording identity		+**46.7**	

Table 9: Sampling-rate sweep under the GROUP protocol (Random Forest, five-seed means). Lower rates match or exceed 100 Hz.

Rate (Hz)	**Raw bitrate (kbps)**	**Accuracy (%)**
100	9.6	52.8
50	4.8	39.2
25	2.4	**62.9**
20	1.9	60.9
12.5	1.2	60.9
10	1.0	58.8

Table 10: Incremental frugal recipe under the GROUP protocol (five-seed means). The resource-optimal node exceeds the naive baseline.

Configuration	RF	KNN	LogReg
Baseline (100 Hz, 6 ch, 2 s)	52.8	59.0	59.6
+ 25 Hz	62.9	46.5	57.3
+ raw 3 ch	65.6	51.4	52.5
+ 4 s window (optimal)	**75.9**	54.9	60.3

Table 11: Leakage-free (GROUP) accuracy of the Random Forest, scored per held-out recording over ten random recording-level partitions. Intervals are 95% intervals for the mean, clustered on recording ($n = 36$); comparisons are two-sided paired permutation tests on the per-recording differences.

Configuration	Accuracy (%)	Macro-F_1 (%)
Baseline (100 Hz, 6 ch, 2 s)	53.9 ± 12.5	49.1
25 Hz, 6 ch, 2 s	63.2 ± 11.6	59.2
Frugal (25 Hz, 3 raw ch, 4 s)	$\mathbf{75.0 \pm 11.4}$	**71.1**
Paired comparison on per-recording differences ($n = 36$)		
Frugal − Baseline	+21.1	$p = 0.020$
25 Hz − Baseline	+9.4	$p = 0.26$ (n.s.)

Table 12: Separating band-limiting, sample count and aliasing (GROUP protocol, Random Forest, five seeds; intervals are 95% clustered on recording, $n = 36$). Conditions (b) and (c) contain identical frequency content and differ only in the number of samples per window.

	Condition	Samples/win.	Accuracy (%)	Macro-F_1 (%)
(a)	100 Hz, unmodified	200	52.8	48.4
(b)	100 Hz, low-pass 12.5 Hz	200	50.1	45.1
(c)	25 Hz, anti-alias decimation	50	**62.9**	**58.6**
(d)	25 Hz, no anti-alias filter	50	38.0	32.2
Attribution of the $+10.1$ *point net change*				
band-limiting alone, (a)→(b)			-2.7	
sample count at fixed band, (b)→(c)			**+12.8**	
aliasing, (c)→(d)			-24.9	

Table 13: Communication link budget: telemetry bitrate versus sustainable uplink capacity. Bitrates assume a 1 s reporting hop, `int16` raw samples and `float32` features; the assumed sustainable application-level uplinks are 300 kbps (Bluetooth Low Energy 4.2), 100 kbps (Zigbee, IEEE 802.15.4), 30 kbps (NB-IoT) and 55 bps (LoRaWAN, a spreading-factor-7 physical rate of 5.47 kbps throttled by a 1% duty cycle). ✓ = sustainable, ✗ = infeasible.

Telemetry stream	Bitrate	BLE	Zigbee	NB-IoT	LoRaWAN
Raw (6 ch, int16)	9.6 kbps	✓	✓	✓	✗
Features (84/s)	2.69 kbps	✓	✓	✓	✗
Features (min., 6/s)	192 bps	✓	✓	✓	✗
On-node decision	40 bps	✓	✓	✓	✓

Table 14: Edge-model footprint versus leakage-free (GROUP) accuracy at the baseline configuration (100 Hz, six channels, 2 s; five-seed means). Serialised sizes are architecture-determined.

Model	GROUP accuracy (%)	Serialised size
Random Forest (200 trees)	52.8	1.09 MB
Random Forest (50, depth 8)	52.9	214 kB
Decision tree (depth 8)	44.3	4.3 kB
Logistic regression	**59.6**	4.7 kB

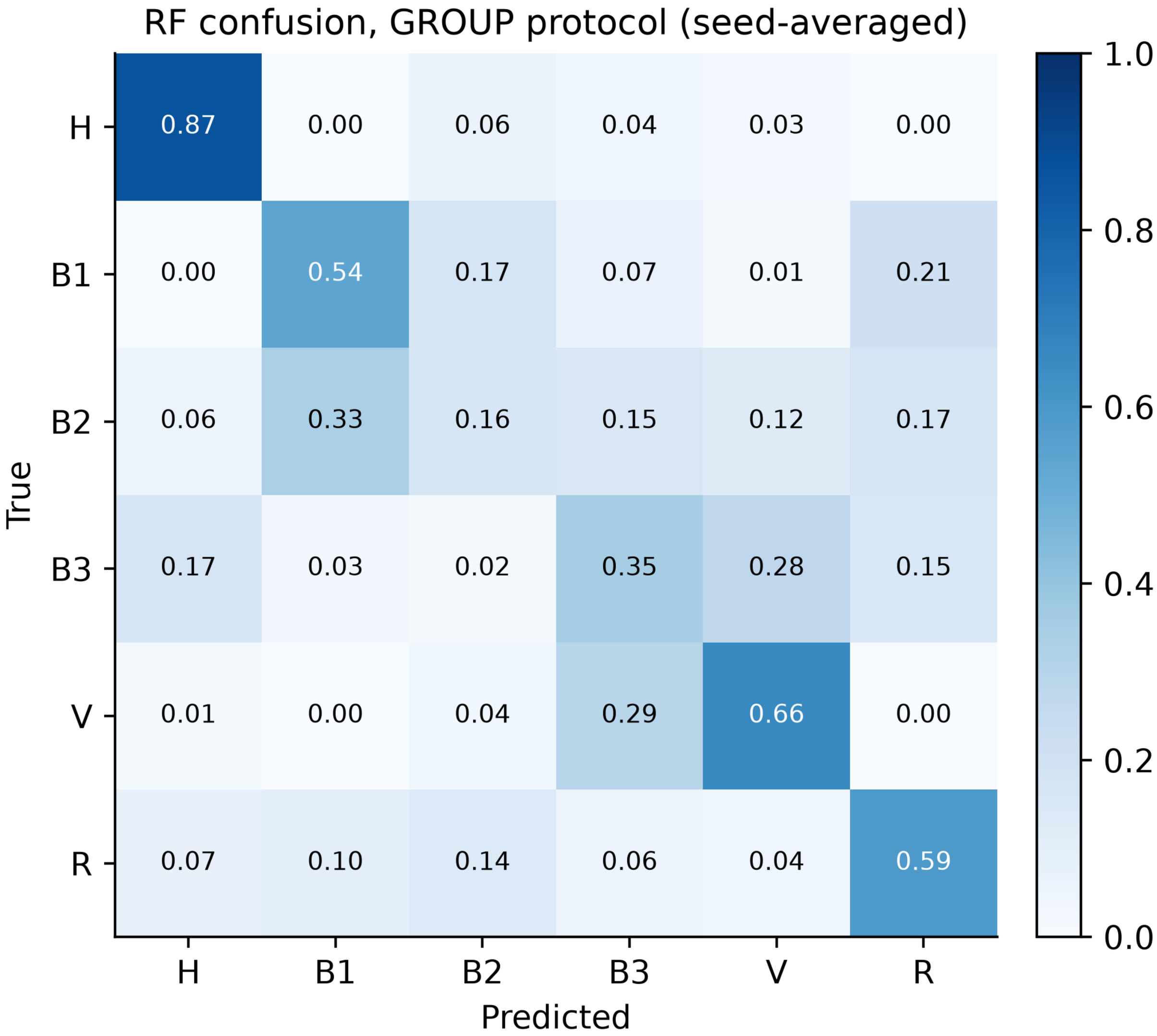

Figure 7: Seed-averaged confusion matrix of the Random Forest under the GROUP protocol (row-normalised; baseline configuration, five recording-level partitions). The healthy state (H) is the most separable, followed by the supply-imbalance (V) and broken-rotor-bar (R) faults; the lubrication-severity states (B1, B2) are the most confused.

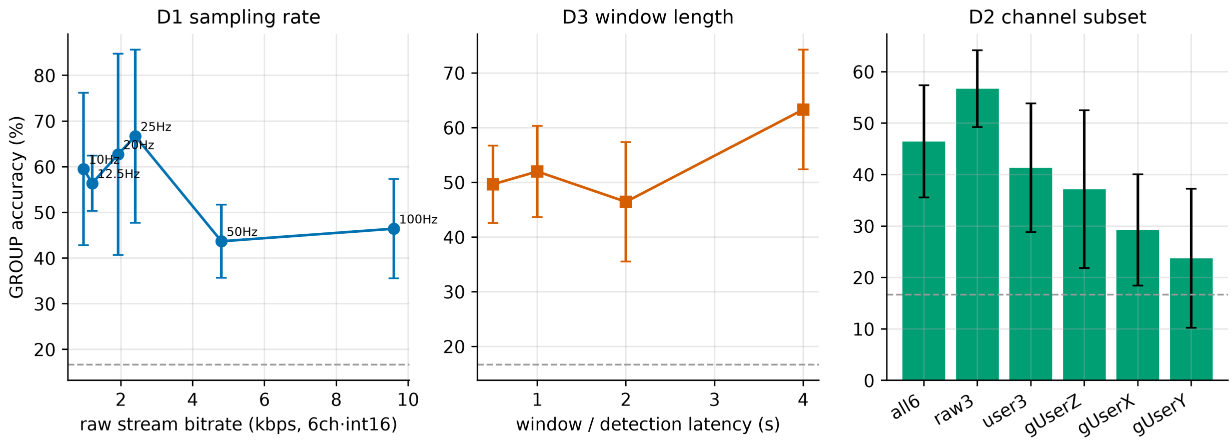

Figure 8: Accuracy–resource trade-offs under the GROUP protocol: sampling rate (left), window length (centre), and channel subset (right). The dashed line marks the 16.7% chance level.

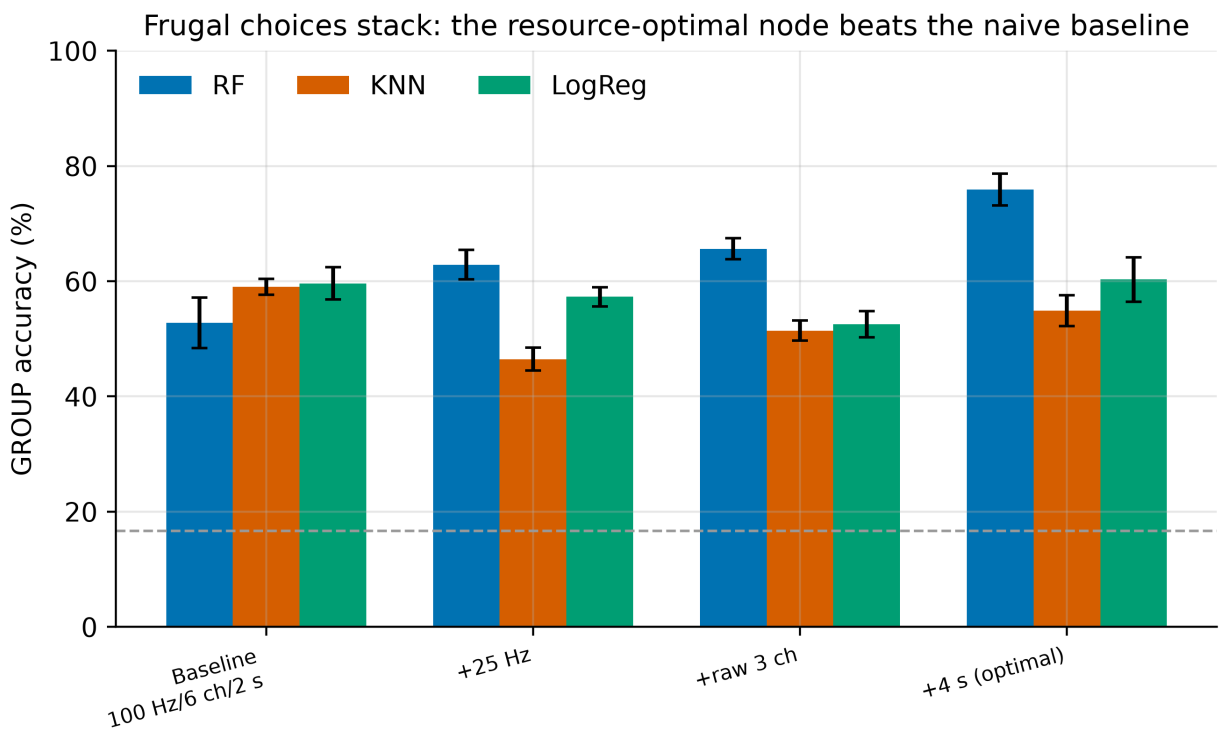

Figure 9: Incremental frugal recipe under the GROUP protocol. Combining the resource-frugal choices raises Random Forest accuracy from 52.8% to 75.9% while reducing the raw data volume roughly eight-fold.